BRCA1 Sequence Variation classification

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Background

Advances in sequencing technologies have enabled researchers to generate billions of nucleotides per day at reduced costs, which leads to producing huge amounts of data very fast (Liu et al. 2012). On the other hand, experimental methods for determining the structure and function of these sequences remain slow and expensive (Thompson et al. 2011). As a result, there is a growing gap between the number of sequences available and our understanding of what they do; that's when computational biology and bioinformatics help fill this gap by predicting structural and functional information from sequence data through sequence alignment methods (Thompson et al. 2011 & Chowdhury and Garai 2017). In computational biology and bioinformatics, sequence alignment is used to align nucleotide sequences (DNA and RNA) or amino acid sequences (protein) to reveal similarities that may reflect shared function or evolutionary history (Langmead and Salzberg 2012). It works by matching identical or similar residues and adds gaps where insertions or deletions may have occurred (Wu and Nacu 2010). In modern genomics, alignment is also used for placing short sequencing reads onto a reference genome, for gene annotation, transcript reconstruction, and identifying splice sites (Chowdhury and Garai 2017). Sequence alignment can be performed as pairwise sequence alignment (PSA), which compares two sequences, or as multiple sequence alignment (MSA), which aligns three or more related sequences, being pairwise the most computationally efficient because it works only with two sequences at a time, while multiple sequence alignment extends this to many sequences, although it helps identify conserved regions that require more time and computational power (Chao et al. 2022). It can also be performed as global alignment, which compares sequences across their full length and works best when sequences are similar and of comparable size, or as local alignment, which focuses on matching only the most similar regions, making it ideal for divergent or long genomic sequences (Chowdhury and Garai 2017). For this project, we are going to work with two multifasta files containing BRCA1 variants, which are among the most clinically important genes because inherited mutations in them significantly increase the risk of breast cancer, especially in women (Semmler et al. 2019). Around 10% of breast cancer cases are hereditary, and BRCA1 mutations alone represent 25% of these cases. BRCA1 is located on chromosome 17 and encodes a large, multidomain protein involved in maintaining genomic stability, particularly through DNA repair pathways such as homologous recombination. Its key domains, including the RING, serine-rich region, and BRCT domains, enable interactions with other proteins essential for repairing damaged DNA (Karami & Mehdipour 2013; Sharma et al. 2018). BRCA1 contains many Alu repeat elements, which make it prone to different types of mutations such as frameshifts, insertions, deletions, missense, and nonsense changes. These alterations often disrupt the gene's reading frame or truncate the protein, preventing BRCA1 from performing its DNA repair functions. When BRCA1 cannot repair DNA properly, genomic instability increases, which contributes to the development of breast cancer and is associated with poorer clinical outcomes (Easton 1999; Sharma et al. 2018). Therefore, in this study, we will assess sequence variation across a set of BRCA1 sequences provided in multifasta format. Through multiple sequence alignment and variant classification, to quantify the types and distribution of mutations present in gene data provided and evaluate their potential biological impact.

Problem question:

- What sequence variations (including SNPs and INDELs) can be identified within a MAFFT-aligned set of BRCA1 sequences, and how those variants could potentially affect the gene?

Aim

- Perform a multiple sequence alignment of BRCA1 sequences isolated and analyse MAFFT-aligned BRCA1 sequences using Python in order to detect, quantify, and compare sequence variations such as SNPs, INDELs, and variable positions.

objectives

1. Perform a multiple sequence alignment of BRCA1 sequences using MAFFT on the command line (Ubuntu Linux).
2. Import and process the aligned FASTA file in Python using Biopython.
3. Perform pairwise sequence similarity between all aligned sequences.
4. Identify variable positions across the alignment.
5. Detect and classify SNPs relative to a reference sequence.
6. Detect and classify INDELs relative to a reference sequence.
7. Interpret the biological significance of the observed sequence variation.

METHODS

Two FASTA files containing the BRCA1 nucleotide sequence were used for this analysis. where the first file consisted of 10 BRCA1 mRNA isolates (almost a full transcript sequence) and the second contained 9 partial gene isolates (specific regions of the gene). because both of the files contained sequences that where diffrent in length and genomic coverage on the BRCA1, they were processed and analysed separately. Both FASTA files were subjected to multiple sequence alignment (MSA) using MAFFT. Following alignment, the resulting aligned FASTA files were imported into Python for downstream computational analysis. Using Biopython and other Python functions, the aligned sequences were examined to identify variable positions, single-nucleotide polymorphisms (SNPs), insertions and deletions (INDELs), and pairwise sequence similarity.

PACKAGES USED

1. SeqIO was used for reading and writing FASTA files.
2. AlignIO was used for reading and handling multiple sequence alignments.
3. SeqRecord was used to store a sequence together with its ID, description, and metadata for downstream computer analysis
4. The subprocess module was used to execute MAFFT directly from Python, enabling automated multiple sequence alignment without manually running MAFFT in a terminal.

important note

Because both FASTA files required the same sequence-processing and analysis steps, the workflow

was written using functions. This avoided repeating the same code for each file; instead, once a function was created, the second file could be processed simply by calling the same function again, which means that first we analysed file 1 and later on we called the functions for file two.

In [1]:
```
# loading the packages that will be used
from Bio import SeqIO
from Bio import AlignIO
from Bio.Seq import Seq
from Bio.SeqRecord import SeqRecord
import subprocess
```

LOADING THE FILES INTO PYTHON

First, each raw FASTA file was loaded into Python and parsed using the SeqIO module. This converted the raw text sequences into structured SeqRecord objects, allowing the sequence data and identifiers to be accessed and visualised. After parsing, the sequences were stored in a dictionary format, with the sequence IDs as keys and the corresponding sequence strings as values for later use. as shown below:

In [ ]:
```
# loading the file 1
BCRA1_file_1 = 'C:/Users/user/Downloads/BRCA1_mRNA_isolate_D.fasta'

## parsing the file
# logic steps
#1) create a function that will parse the file
#2) read the file
#3)create empty strings form my header and sequences
#4)define whats a header is and what a sequence is
#3) return a dictionary

from Bio import SeqIO

def parse_file(filename, format):
    '''Reads a FASTA file using Biopython and returns a dictionary
       where the key is the sequence ID and the value is the sequence.'''

    Seq_dict = {}   # dictionary to store sequences

    for record in SeqIO.parse(filename, format):
        header = record.id
        sequence = str(record.seq)
        Seq_dict[header] = sequence

    return Seq_dict



# parsing the file 1
BCRA1_file_1 = 'C:/Users/user/Downloads/BRCA1_mRNA_isolate_D.fasta'
print(parse_file(BCRA1_file_1, 'fasta'))
```

PREPARING THE SEQUENCE FOR ALIGNMENT ON MAFFT

The stored sequences in the dictionary were then converted into a list and written to a temporary FASTA file. This ensured that the sequences were formatted correctly and consistently before being passed into the alignment step, as presented below:

In [4]:
```
# Preparing sequences for alignment
## logic steps
#1)Create an empty list to store SeqRecord objects
```

```
#2)Loop through the dictionary items (header and sequence)
#3)Convert each sequence back into a seqrecord
#4)Add each SeqRecord to the list
#5)This list will be written to a FASTA file for alignment

def seq_records(data):
    '''
    Converts a dictionary of sequences into a list of SeqRecord objects. for later u
    '''
    seq_records = []
    for header, sequence in data.items():
        rec = SeqRecord(Seq(sequence), id=header, description="")
        seq_records.append(rec)
    return seq_records

def write_fasta(seq_records, outfile):
    '''
    Writes a list of SeqRecord objects to a FASTA file and returns a temporary fasta
    '''
    SeqIO.write(seq_records, outfile, "fasta")
    return outfile

data1 = parse_file(BCRA1_file_1, 'fasta')
records_file_1 = seq_records(data1)
print(records_file_1)

temporary_file_1 = write_fasta(records_file_1, "temp1.fasta")
print(temporary_file_1)
```

Multiple Sequence alignment (MSA)

Global Multiple sequence alignment was performed outside Python using the Ubuntu command line. MAFFT was first installed in the Ubuntu environment to ensure that the alignment could be executed from the command line. Once the sequences had been parsed and written into a temporary FASTA files (temp1 for the BRCA1 mRNA isolates and temp2 for partial gene isolates), these temporary files were passed to MAFFT for alignment directly in Ubuntu using the MAFFT command, which generated a new aligned FASTA file containing the sequences arranged in a consistent positional framework. Once the alignment was complete, the aligned FASTA file was reloaded into Python as follows below:

In [ ]:
```
# performing Global MSA alignment using MAFFT

## logic steps
#1) Take the fasta file path as input
#2) change the windows path into a linux path
#3) create the command that will run mafft inside the Linux command line
#4) run the command and capture the aligned file
#5) return the aligned sequences as text so I can save or use them later

import subprocess

def align_with_mafft(input_fasta):
    '''Runs MAFFT inside Ubuntu to perform a global multiple sequence alignment.Conv
    command-line-compatible path. Executes MAFFT using subprocess
    Captures and returns the aligned FASTA file as a string'''

    # Convert Windows path to Ubuntu command line path
    ubunto_in = input_fasta.replace("C:\\", "/mnt/c/").replace("\\", "/")

    # Command to run MAFFT inside Ubuntu command line path
    cmd = ["wsl", "mafft", ubunto_in]

    # Run MAFFT and capture output
    result = subprocess.run(cmd, capture_output=True, text=True)
```

```
    # Return aligned sequences
    return result.stdout

aligned_file1 = align_with_mafft("C:\\Users\\user\\Downloads\\temp1.fasta")
print(aligned_file1)
```

DOWNSTREAM COMPUTER ANALYSIs

ANALISYS OF DATASET 1 #(BRCA1 mRNA iISOLATES) AND RESULTS

1. Similarity score between sequences

The downstream analysis started with the calculation of the pairwise similarity between the aligned sequences. This was done by comparing each sequence position by position and computing the percentage of matching nucleotides with the purpose of obtaining a general measure of how related the sequences were before examining specific types of variation.

The BRCA1 mRNA sequences in File 1 were similar to each other, with most pairs showing between 97% and 98% identity. For example, MF590182.1_cds_AWA45334.1_1 shared 98% similarity with both MF590181.1_cds_AWA45333.1_1 and MF590180.1_cds_AWA45332.1_1. The highest similarity in the dataset was 99%, observed between MF590180.1_cds_AWA45332.1_1 and MF590174.1_cds_AWA45326.1_1, showing that these two isolates are almost identical. The lowest value was 96%, found between MF590181.1_cds_AWA45333.1_1 and MF590173.1_cds_AWA45325.1_1, but even this still indicates strong conservation or similarity of the sequences confirmed by the phylogenetic tree (Figure 1).

In [9]:
```
## Read the aligned FASTA file
## logic steps
#1) create a function that will read the aligned fasta file
#2) open the file using SeqIO.parse
#3) convert the parsed data into a list of SeqRecord objects
#4) return this list so I can use it for SNPs, indels, similarity etc

def read_alignment(path):
    ''' Reads an aligned FASTA file and returns a list of SeqRecord objects and retu
    '''
    records = list(SeqIO.parse(path, "fasta"))
    return records
records = read_alignment("C:\\Users\\user\\Downloads\\aligned1.fasta")
```

In [11]:
```
# LOGIC STEPS
# 1) create a function and Start a counter at zero to track the current position
# 2) Start a match counter at zero to count identical nucleotides
# 3) loop through every position in the sequences
# 4) Check if the nucleotides at the same position are equal
# 5) If they match and are not gaps, count
# 6) move to the next position until the end of the sequence
# 7) Calculate the per cent identity
# 8) return the percentage as an integer

def similarity(seq1, seq2):
    '''Calculates the percentage similarity between two aligned sequences
    by counting how many positions contain the same nucleotide'''

    count = 0
    matches = 0
```

```
    while count < len(seq1):
        if seq1[count] == seq2[count] and seq1[count] != "-":
            matches = matches + 1
        count = count + 1
    percent = int((matches / len(seq1)) * 100)
    return percent

## logic steps
#1) create a function that will compare every sequence with every other sequence
#2) get how many sequences exist (n = number of sequences)
#3) use the first loop to pick the first sequence in the pair
#4) use the second loop to pick the second sequence in the pair
#5) convert both sequences to strings
#6) calculate percent identity using the similarity() function
#7) print the result for each pair

def compare_all(records):
    '''
    Compares every sequence with every other sequence in the records by looping thro
    all unique sequence pairs and prints the percent identity.
    '''

    n = len(records)
    count = 0

    while count < n:
        count_1 = count + 1

        while count_1 < n:
            seq1 = str(records[count].seq)
            seq2 = str(records[count_1].seq)

            score = similarity(seq1, seq2)

            print(records[count].id, "vs", records[count_1].id, "=", score, "%")

            count_1 = count_1 + 1

        count = count + 1
```

In [13]:
```
compare_all(records)
```

```
lcl|MF590182.1_cds_AWA45334.1_1 vs lcl|MF590181.1_cds_AWA45333.1_1 = 98 %
lcl|MF590182.1_cds_AWA45334.1_1 vs lcl|MF590180.1_cds_AWA45332.1_1 = 98 %
lcl|MF590182.1_cds_AWA45334.1_1 vs lcl|MF590179.1_cds_AWA45331.1_1 = 97 %
lcl|MF590182.1_cds_AWA45334.1_1 vs lcl|MF590178.1_cds_AWA45330.1_1 = 98 %
lcl|MF590182.1_cds_AWA45334.1_1 vs lcl|MF590177.1_cds_AWA45329.1_1 = 97 %
lcl|MF590182.1_cds_AWA45334.1_1 vs lcl|MF590176.1_cds_AWA45328.1_1 = 97 %
lcl|MF590182.1_cds_AWA45334.1_1 vs lcl|MF590175.1_cds_AWA45327.1_1 = 98 %
lcl|MF590182.1_cds_AWA45334.1_1 vs lcl|MF590174.1_cds_AWA45326.1_1 = 97 %
lcl|MF590182.1_cds_AWA45334.1_1 vs lcl|MF590173.1_cds_AWA45325.1_1 = 97 %
lcl|MF590181.1_cds_AWA45333.1_1 vs lcl|MF590180.1_cds_AWA45332.1_1 = 98 %
lcl|MF590181.1_cds_AWA45333.1_1 vs lcl|MF590179.1_cds_AWA45331.1_1 = 97 %
lcl|MF590181.1_cds_AWA45333.1_1 vs lcl|MF590178.1_cds_AWA45330.1_1 = 97 %
lcl|MF590181.1_cds_AWA45333.1_1 vs lcl|MF590177.1_cds_AWA45329.1_1 = 97 %
lcl|MF590181.1_cds_AWA45333.1_1 vs lcl|MF590176.1_cds_AWA45328.1_1 = 97 %
lcl|MF590181.1_cds_AWA45333.1_1 vs lcl|MF590175.1_cds_AWA45327.1_1 = 98 %
lcl|MF590181.1_cds_AWA45333.1_1 vs lcl|MF590174.1_cds_AWA45326.1_1 = 97 %
lcl|MF590181.1_cds_AWA45333.1_1 vs lcl|MF590173.1_cds_AWA45325.1_1 = 96 %
lcl|MF590180.1_cds_AWA45332.1_1 vs lcl|MF590179.1_cds_AWA45331.1_1 = 97 %
lcl|MF590180.1_cds_AWA45332.1_1 vs lcl|MF590178.1_cds_AWA45330.1_1 = 98 %
lcl|MF590180.1_cds_AWA45332.1_1 vs lcl|MF590177.1_cds_AWA45329.1_1 = 97 %
lcl|MF590180.1_cds_AWA45332.1_1 vs lcl|MF590176.1_cds_AWA45328.1_1 = 97 %
lcl|MF590180.1_cds_AWA45332.1_1 vs lcl|MF590175.1_cds_AWA45327.1_1 = 98 %
lcl|MF590180.1_cds_AWA45332.1_1 vs lcl|MF590174.1_cds_AWA45326.1_1 = 99 %
lcl|MF590180.1_cds_AWA45332.1_1 vs lcl|MF590173.1_cds_AWA45325.1_1 = 97 %
lcl|MF590179.1_cds_AWA45331.1_1 vs lcl|MF590178.1_cds_AWA45330.1_1 = 97 %
lcl|MF590179.1_cds_AWA45331.1_1 vs lcl|MF590177.1_cds_AWA45329.1_1 = 97 %
lcl|MF590179.1_cds_AWA45331.1_1 vs lcl|MF590176.1_cds_AWA45328.1_1 = 97 %
lcl|MF590179.1_cds_AWA45331.1_1 vs lcl|MF590175.1_cds_AWA45327.1_1 = 97 %
lcl|MF590179.1_cds_AWA45331.1_1 vs lcl|MF590174.1_cds_AWA45326.1_1 = 97 %
lcl|MF590179.1_cds_AWA45331.1_1 vs lcl|MF590173.1_cds_AWA45325.1_1 = 97 %
lcl|MF590178.1_cds_AWA45330.1_1 vs lcl|MF590177.1_cds_AWA45329.1_1 = 97 %
lcl|MF590178.1_cds_AWA45330.1_1 vs lcl|MF590176.1_cds_AWA45328.1_1 = 97 %
lcl|MF590178.1_cds_AWA45330.1_1 vs lcl|MF590175.1_cds_AWA45327.1_1 = 98 %
lcl|MF590178.1_cds_AWA45330.1_1 vs lcl|MF590174.1_cds_AWA45326.1_1 = 97 %
lcl|MF590178.1_cds_AWA45330.1_1 vs lcl|MF590173.1_cds_AWA45325.1_1 = 97 %
lcl|MF590177.1_cds_AWA45329.1_1 vs lcl|MF590176.1_cds_AWA45328.1_1 = 97 %
lcl|MF590177.1_cds_AWA45329.1_1 vs lcl|MF590175.1_cds_AWA45327.1_1 = 97 %
lcl|MF590177.1_cds_AWA45329.1_1 vs lcl|MF590174.1_cds_AWA45326.1_1 = 97 %
lcl|MF590177.1_cds_AWA45329.1_1 vs lcl|MF590173.1_cds_AWA45325.1_1 = 97 %
lcl|MF590176.1_cds_AWA45328.1_1 vs lcl|MF590175.1_cds_AWA45327.1_1 = 97 %
lcl|MF590176.1_cds_AWA45328.1_1 vs lcl|MF590174.1_cds_AWA45326.1_1 = 97 %
lcl|MF590176.1_cds_AWA45328.1_1 vs lcl|MF590173.1_cds_AWA45325.1_1 = 97 %
lcl|MF590175.1_cds_AWA45327.1_1 vs lcl|MF590174.1_cds_AWA45326.1_1 = 97 %
lcl|MF590175.1_cds_AWA45327.1_1 vs lcl|MF590173.1_cds_AWA45325.1_1 = 97 %
lcl|MF590174.1_cds_AWA45326.1_1 vs lcl|MF590173.1_cds_AWA45325.1_1 = 97 %
```

1.1 Phylogenetic tree

The phylogenetic tree was constructed using pairwise sequence similarity values and clustered with the UPGMA method to illustrate the relationships among the BRCA1 mRNA isolates. All sequences form a grouped cluster, reflecting the high degree of conservation observed across the dataset. Branch lengths are short and uniform, indicating minimal genetic divergence among isolates and supporting the interpretation that these sequences represent closely related variants of the same BRCA1 coding region (Figure 2).

In [58]:
```python
import numpy as np
from scipy.cluster.hierarchy import dendrogram, linkage
from scipy.spatial.distance import squareform
import matplotlib.pyplot as plt


def build_phylogenetic_tree(similarity_matrix, names, title="Phylogenetic Tree"):
    '''
    Builds a simple UPGMA phylogenetic tree using a similarity matrix (% identity).
```

```
    where similarity_matrix is  NxN numpy array with similarity values (0–100)
    and names is  list of sequence names (length N)
    '''

    # Convert similarity (%) to distance (0 = identical, 1 = completely different)
    distance_matrix = 1 - (similarity_matrix / 100)

    # Convert square matrix to condensed form for SciPy
    condensed = squareform(distance_matrix)

    # Build the tree using UPGMA (average linkage)
    Z = linkage(condensed, method='average')

    # Plot the dendrogram
    plt.figure(figsize=(12, 6))
    dendrogram(Z, labels=names, leaf_rotation=90)
    plt.title(title)
    plt.ylabel("Distance")
    plt.tight_layout()
    plt.show()

    #return Z

names_file1 = [
    "MF590182.1",
    "MF590181.1",
    "MF590180.1",
    "MF590179.1",
    "MF590178.1",
    "MF590177.1",
    "MF590176.1",
    "MF590175.1",
    "MF590174.1",
    "MF590173.1"
]


similarity_file1 = np.array([
    [100, 98, 98, 97, 98, 97, 97, 98, 97, 97],
    [ 98,100, 98, 97, 97, 97, 97, 98, 97, 96],
    [ 98, 98,100, 97, 98, 97, 97, 98, 99, 97],
    [ 97, 97, 97,100, 97, 97, 97, 97, 97, 97],
    [ 98, 97, 98, 97,100, 97, 97, 98, 97, 97],
    [ 97, 97, 97, 97, 97,100, 97, 97, 97, 97],
    [ 97, 97, 97, 97, 97, 97,100, 97, 97, 97],
    [ 98, 98, 98, 97, 98, 97, 97,100, 97, 97],
    [ 97, 97, 99, 97, 97, 97, 97, 97,100, 97],
    [ 97, 96, 97, 97, 97, 97, 97, 97, 97,100]
])


build_phylogenetic_tree(
    similarity_file1,
    names_file1,
    title="Phylogenetic Tree – File 1 (Full BRCA1 mRNA)"
)
```

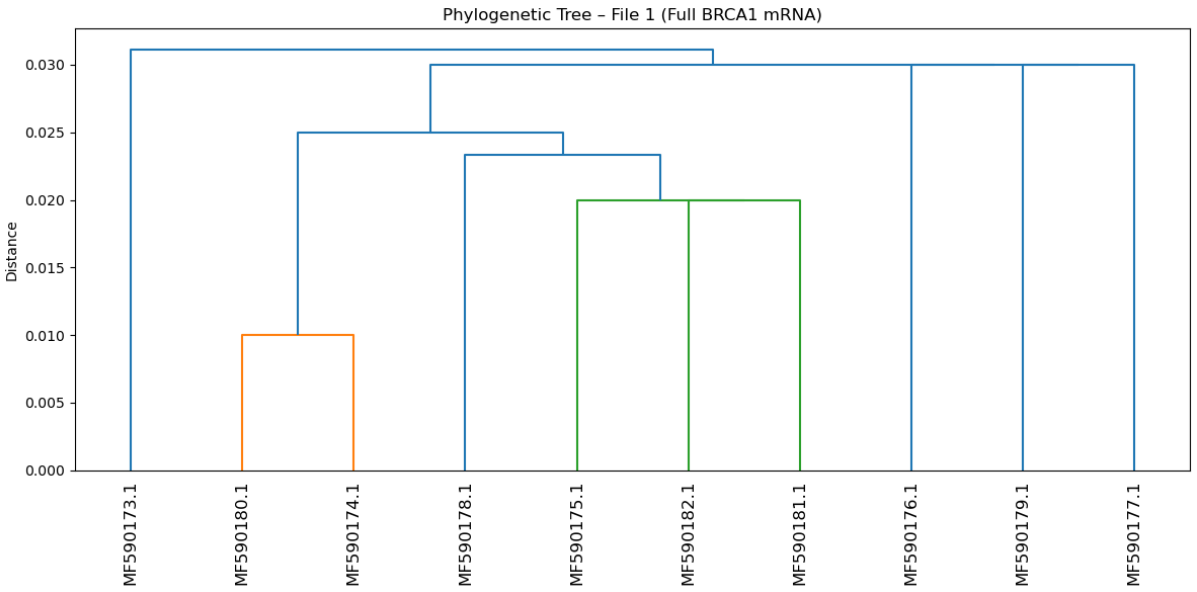


Figure 1- Phylogenetic tree of BRCA1 full-length mRNA isolates (File 1)

2. Differences between the sequences

After obtaining the similarity percentages, the next step involved identifying the exact positions where the sequences differed by scanning the alignment and recording every site where the nucleotides were not identical across sequences. as shown bellow:

2.1. Identifying SNPs (pairwise)

In this project, a true polymorphism was defined as a base-to-base substitution where both sequences contain a nucleotide at the same position. Differences involving gaps were not considered SNPs, as they represent insertions or deletions rather than single-nucleotide changes (Marwal & Gaur, 2020). The SNPs identification in the BRCA1 sequences was performed by comparing every sequence against every other sequence using a pairwise approach.

The SNP analysis of the BRCA1 mRNA sequences in File 1 showed that almost all of the sequences are identical at every position where both sequences contain a base. Only one substitution was observed across the dataset. This SNP occurred at position 161, where some sequences have a **g**, and others have a **c**. For example, when comparing MF590182.1_cds_AWA45334.1_1 with MF590181.1_cds_AWA45333.1_1, the only difference between them is this single base change at position 161 (from g to c). The same substitution also appears when MF590181.1 is compared with several other sequences, including MF590179.1, MF590178.1, MF590177.1, and MF590176.1, showing that this SNP is shared by a group of isolates. No other base-to-base substitutions were detected in the rest of the alignment. All remaining differences in the raw output seemed to be caused by gaps or missing bases, which are not SNPs.

In [14]:
```
# LOGIC STEPS
# 1) Create a counting variable
# 2) loop through every position in the sequences
# 3) Check if the nucleotides at the same position are different
# 4) If they differ, print the position and the two nucleotides
# 5) move to the next position until the end of the sequence

def show_differences(seq1, seq2):
    '''
```

```
    Compares two aligned sequences and prints the positions
    where the nucleotides are different.
    '''

    count = 0
    while count < len(seq1):
        if seq1[count] != seq2[count]:
            print("Position", count + 1, ":", seq1[count], "vs", seq2[count])
        count = count + 1

## logic steps
#1) create a function that will compare all sequences pairwise
#2) get how many sequences exist (n = number of sequences)
#3) use the first loop to pick the first sequence in the pair
#4) use the second loop to pick the second sequence in the pair
#5) print which two sequences are being compared
#6) convert both sequences to strings
#7) call show_differences() to print all mismatches
#8) move to the next pair until all pairs are compared


def show_all_differences(records):
    '''
    Compares all sequences pairwise and prints all differences
    for each pair using show_differences().
    '''

    n = len(records)
    count = 0

    while count < n:
        count_1 = count + 1

        while count_1 < n:
            print("\nComparing:", records[count].id, "vs", records[count_1].id)

            seq1 = str(records[count].seq)
            seq2 = str(records[count_1].seq)

            show_differences(seq1, seq2)

            count_1 = count_1 + 1

        count = count + 1
```

In [41]:
```
show_all_differences(records)
```

```
Comparing: lcl|MF590182.1_cds_AWA45334.1_1 vs lcl|MF590181.1_cds_AWA45333.1_1
Position 161 : g vs c

Comparing: lcl|MF590182.1_cds_AWA45334.1_1 vs lcl|MF590180.1_cds_AWA45332.1_1
Position 1 : - vs g
Position 2 : - vs a
Position 3 : - vs g

Comparing: lcl|MF590182.1_cds_AWA45334.1_1 vs lcl|MF590179.1_cds_AWA45331.1_1
Position 240 : c vs -
Position 241 : a vs -

Comparing: lcl|MF590182.1_cds_AWA45334.1_1 vs lcl|MF590178.1_cds_AWA45330.1_1
Position 241 : a vs -

Comparing: lcl|MF590182.1_cds_AWA45334.1_1 vs lcl|MF590177.1_cds_AWA45329.1_1
Position 240 : c vs -
Position 241 : a vs -

Comparing: lcl|MF590182.1_cds_AWA45334.1_1 vs lcl|MF590176.1_cds_AWA45328.1_1
Position 239 : c vs -
Position 240 : c vs -
Position 241 : a vs -

Comparing: lcl|MF590182.1_cds_AWA45334.1_1 vs lcl|MF590175.1_cds_AWA45327.1_1

Comparing: lcl|MF590182.1_cds_AWA45334.1_1 vs lcl|MF590174.1_cds_AWA45326.1_1
Position 1 : - vs g
Position 2 : - vs a
Position 3 : - vs g
Position 240 : c vs -
Position 241 : a vs -

Comparing: lcl|MF590182.1_cds_AWA45334.1_1 vs lcl|MF590173.1_cds_AWA45325.1_1
Position 238 : c vs -
Position 239 : c vs -
Position 240 : c vs -
Position 241 : a vs -

Comparing: lcl|MF590181.1_cds_AWA45333.1_1 vs lcl|MF590180.1_cds_AWA45332.1_1
Position 1 : - vs g
Position 2 : - vs a
Position 3 : - vs g
Position 161 : c vs g

Comparing: lcl|MF590181.1_cds_AWA45333.1_1 vs lcl|MF590179.1_cds_AWA45331.1_1
Position 161 : c vs g
Position 240 : c vs -
Position 241 : a vs -

Comparing: lcl|MF590181.1_cds_AWA45333.1_1 vs lcl|MF590178.1_cds_AWA45330.1_1
Position 161 : c vs g
Position 241 : a vs -

Comparing: lcl|MF590181.1_cds_AWA45333.1_1 vs lcl|MF590177.1_cds_AWA45329.1_1
Position 161 : c vs g
Position 240 : c vs -
Position 241 : a vs -

Comparing: lcl|MF590181.1_cds_AWA45333.1_1 vs lcl|MF590176.1_cds_AWA45328.1_1
Position 161 : c vs g
Position 239 : c vs -
Position 240 : c vs -
Position 241 : a vs -

Comparing: lcl|MF590181.1_cds_AWA45333.1_1 vs lcl|MF590175.1_cds_AWA45327.1_1
Position 161 : c vs g
```

```
Comparing: lcl|MF590181.1_cds_AWA45333.1_1 vs lcl|MF590174.1_cds_AWA45326.1_1
Position 1 : - vs g
Position 2 : - vs a
Position 3 : - vs g
Position 161 : c vs g
Position 240 : c vs -
Position 241 : a vs -

Comparing: lcl|MF590181.1_cds_AWA45333.1_1 vs lcl|MF590173.1_cds_AWA45325.1_1
Position 161 : c vs g
Position 238 : c vs -
Position 239 : c vs -
Position 240 : c vs -
Position 241 : a vs -

Comparing: lcl|MF590180.1_cds_AWA45332.1_1 vs lcl|MF590179.1_cds_AWA45331.1_1
Position 1 : g vs -
Position 2 : a vs -
Position 3 : g vs -
Position 240 : c vs -
Position 241 : a vs -

Comparing: lcl|MF590180.1_cds_AWA45332.1_1 vs lcl|MF590178.1_cds_AWA45330.1_1
Position 1 : g vs -
Position 2 : a vs -
Position 3 : g vs -
Position 241 : a vs -

Comparing: lcl|MF590180.1_cds_AWA45332.1_1 vs lcl|MF590177.1_cds_AWA45329.1_1
Position 1 : g vs -
Position 2 : a vs -
Position 3 : g vs -
Position 240 : c vs -
Position 241 : a vs -

Comparing: lcl|MF590180.1_cds_AWA45332.1_1 vs lcl|MF590176.1_cds_AWA45328.1_1
Position 1 : g vs -
Position 2 : a vs -
Position 3 : g vs -
Position 239 : c vs -
Position 240 : c vs -
Position 241 : a vs -

Comparing: lcl|MF590180.1_cds_AWA45332.1_1 vs lcl|MF590175.1_cds_AWA45327.1_1
Position 1 : g vs -
Position 2 : a vs -
Position 3 : g vs -

Comparing: lcl|MF590180.1_cds_AWA45332.1_1 vs lcl|MF590174.1_cds_AWA45326.1_1
Position 240 : c vs -
Position 241 : a vs -

Comparing: lcl|MF590180.1_cds_AWA45332.1_1 vs lcl|MF590173.1_cds_AWA45325.1_1
Position 1 : g vs -
Position 2 : a vs -
Position 3 : g vs -
Position 238 : c vs -
Position 239 : c vs -
Position 240 : c vs -
Position 241 : a vs -

Comparing: lcl|MF590179.1_cds_AWA45331.1_1 vs lcl|MF590178.1_cds_AWA45330.1_1
Position 240 : - vs c

Comparing: lcl|MF590179.1_cds_AWA45331.1_1 vs lcl|MF590177.1_cds_AWA45329.1_1
```

```
Comparing: lcl|MF590179.1_cds_AWA45331.1_1 vs lcl|MF590176.1_cds_AWA45328.1_1
Position 239 : c vs -

Comparing: lcl|MF590179.1_cds_AWA45331.1_1 vs lcl|MF590175.1_cds_AWA45327.1_1
Position 240 : - vs c
Position 241 : - vs a

Comparing: lcl|MF590179.1_cds_AWA45331.1_1 vs lcl|MF590174.1_cds_AWA45326.1_1
Position 1 : - vs g
Position 2 : - vs a
Position 3 : - vs g

Comparing: lcl|MF590179.1_cds_AWA45331.1_1 vs lcl|MF590173.1_cds_AWA45325.1_1
Position 238 : c vs -
Position 239 : c vs -

Comparing: lcl|MF590178.1_cds_AWA45330.1_1 vs lcl|MF590177.1_cds_AWA45329.1_1
Position 240 : c vs -

Comparing: lcl|MF590178.1_cds_AWA45330.1_1 vs lcl|MF590176.1_cds_AWA45328.1_1
Position 239 : c vs -
Position 240 : c vs -

Comparing: lcl|MF590178.1_cds_AWA45330.1_1 vs lcl|MF590175.1_cds_AWA45327.1_1
Position 241 : - vs a

Comparing: lcl|MF590178.1_cds_AWA45330.1_1 vs lcl|MF590174.1_cds_AWA45326.1_1
Position 1 : - vs g
Position 2 : - vs a
Position 3 : - vs g
Position 240 : c vs -

Comparing: lcl|MF590178.1_cds_AWA45330.1_1 vs lcl|MF590173.1_cds_AWA45325.1_1
Position 238 : c vs -
Position 239 : c vs -
Position 240 : c vs -

Comparing: lcl|MF590177.1_cds_AWA45329.1_1 vs lcl|MF590176.1_cds_AWA45328.1_1
Position 239 : c vs -

Comparing: lcl|MF590177.1_cds_AWA45329.1_1 vs lcl|MF590175.1_cds_AWA45327.1_1
Position 240 : - vs c
Position 241 : - vs a

Comparing: lcl|MF590177.1_cds_AWA45329.1_1 vs lcl|MF590174.1_cds_AWA45326.1_1
Position 1 : - vs g
Position 2 : - vs a
Position 3 : - vs g

Comparing: lcl|MF590177.1_cds_AWA45329.1_1 vs lcl|MF590173.1_cds_AWA45325.1_1
Position 238 : c vs -
Position 239 : c vs -

Comparing: lcl|MF590176.1_cds_AWA45328.1_1 vs lcl|MF590175.1_cds_AWA45327.1_1
Position 239 : - vs c
Position 240 : - vs c
Position 241 : - vs a

Comparing: lcl|MF590176.1_cds_AWA45328.1_1 vs lcl|MF590174.1_cds_AWA45326.1_1
Position 1 : - vs g
Position 2 : - vs a
Position 3 : - vs g
Position 239 : - vs c

Comparing: lcl|MF590176.1_cds_AWA45328.1_1 vs lcl|MF590173.1_cds_AWA45325.1_1
Position 238 : c vs -
```

```
Comparing: lcl|MF590175.1_cds_AWA45327.1_1 vs lcl|MF590174.1_cds_AWA45326.1_1
Position 1 : - vs g
Position 2 : - vs a
Position 3 : - vs g
Position 240 : c vs -
Position 241 : a vs -

Comparing: lcl|MF590175.1_cds_AWA45327.1_1 vs lcl|MF590173.1_cds_AWA45325.1_1
Position 238 : c vs -
Position 239 : c vs -
Position 240 : c vs -
Position 241 : a vs -

Comparing: lcl|MF590174.1_cds_AWA45326.1_1 vs lcl|MF590173.1_cds_AWA45325.1_1
Position 1 : g vs -
Position 2 : a vs -
Position 3 : g vs -
Position 238 : c vs -
Position 239 : c vs -
```

3. Identifying SNPs relative to the reference sequence (MF590182.1_cds_AWA45334.1_1)

The single-nucleotide polymorphisms (SNPs) were identified by using the first sequence in the alignment (MF590182.1_cds_AWA45334.1_1) as a reference (wild-type), and each remaining sequence was compared against it. Positions where both sequences contained nucleotides (not gaps) but differed in identity were classified as SNPs. The reason why the first sequence was selected as wildtype is that it showed zero SNPs in the pairwise analysis and therefore represents the consensus sequence of the dataset. All BRCA1 sequences in File 1 were completely identical to the reference sequence MF590182.1_cds_AWA45334.1_1. When compared to the reference, nine out of the ten sequences showed zero SNPs but one sequence, MF590181.1_cds_AWA45333.1_1, carried a single true SNP. This substitution occurred at position 161, where the reference sequence has a “g”, and MF590181.1 has a “c”.

In [43]:
```python
# LOGIC STEPS
# 1) Create a count variable starting at zero
# 2) loop through every position in the reference sequence
# 3) compare the nucleotide in seq_ref with the nucleotide in seq_other
# 4) If the two nucleotides are different, increase the SNP counter
# 5) move to the next position until the end of the sequence is reached
# 6) Return the total number of SNPs found

def count_snps(seq_ref, seq_other):
    '''
    Counts how many SNPs exist between a reference sequence
    and another aligned sequence, ignoring gaps.
    '''

    count = 0
    snps = 0

    while count < len(seq_ref):
        if seq_ref[count] != "-" and seq_other[count] != "-":
            if seq_ref[count] != seq_other[count]:
                snps = snps + 1
        count = count + 1

    return snps
```

In [18]:
```python
 # LOGIC STEPS
# 1) choose the first sequence as reference
# 2) loop through all sequences
```

```
# 3) call count_snps(ref, seq)
# 4) print SNP count for each sequence

def snps_per_sequence(records):
    '''
    Prints the number of SNPs in each sequence relative to the first sequence.
    '''

    ref = str(records[0].seq)
    n = len(records)

    print("SNP counts relative to reference:", records[0].id, "\n")

    count = 0
    while count < n:
        seq = str(records[count].seq)
        snps = count_snps(ref, seq)
        print(f"{records[count].id} → {snps} SNPs")
        count = count + 1
```

In [22]:

```
## Logic steps:
#1) Select the first sequence as the reference.
#2) Convert the reference to a string.
#3) Count how many sequences exist.
#4) Start a loop to go through each sequence.
#5) Convert the current sequence to a string.
#6) Start a position counter at 0.
#7) Loop through every position in the sequences.
#8) Check if both positions have real nucleotides (no gaps).
#9) Check if the nucleotides are different.
#10) If different, print the SNP position and bases.
#11) Move to the next sequence until all are processed.
def identify_snps(records):
    '''
    Identifies SNPs in each sequence relative to the first sequence.
    '''
    ref = str(records[0].seq)
    n = len(records)

    print("Reference sequence:", records[0].id, "\n")

    count = 0
    while count < n:
        seq = str(records[count].seq)
        print(records[count].id)

        pos = 0
        while pos < len(ref):
            if ref[pos] != seq[pos] and ref[pos] != "-" and seq[pos] != "-":
                print("  Position", pos + 1, ":", ref[pos], "→", seq[pos])
            pos = pos + 1

        print()
        count = count + 1
```

In [19]:

```
snps_per_sequence(records)
```

```
SNP counts relative to reference: lcl|MF590182.1_cds_AWA45334.1_1

lcl|MF590182.1_cds_AWA45334.1_1 → 0 SNPs
lcl|MF590181.1_cds_AWA45333.1_1 → 1 SNPs
lcl|MF590180.1_cds_AWA45332.1_1 → 0 SNPs
lcl|MF590179.1_cds_AWA45331.1_1 → 0 SNPs
lcl|MF590178.1_cds_AWA45330.1_1 → 0 SNPs
lcl|MF590177.1_cds_AWA45329.1_1 → 0 SNPs
lcl|MF590176.1_cds_AWA45328.1_1 → 0 SNPs
lcl|MF590175.1_cds_AWA45327.1_1 → 0 SNPs
lcl|MF590174.1_cds_AWA45326.1_1 → 0 SNPs
lcl|MF590173.1_cds_AWA45325.1_1 → 0 SNPs
```

In [23]:
```
identify_snps(records)
```

```
Reference sequence: lcl|MF590182.1_cds_AWA45334.1_1

lcl|MF590182.1_cds_AWA45334.1_1

lcl|MF590181.1_cds_AWA45333.1_1
  Position 161 : g → c

lcl|MF590180.1_cds_AWA45332.1_1

lcl|MF590179.1_cds_AWA45331.1_1

lcl|MF590178.1_cds_AWA45330.1_1

lcl|MF590177.1_cds_AWA45329.1_1

lcl|MF590176.1_cds_AWA45328.1_1

lcl|MF590175.1_cds_AWA45327.1_1

lcl|MF590174.1_cds_AWA45326.1_1

lcl|MF590173.1_cds_AWA45325.1_1
```

3.1 Identifying INDELs

Deletions were identified by considering positions where one sequence contained a gap while the reference sequence contained a nucleotide, and insertions by examining positions where one sequence contained a nucleotide while the reference sequence contained a gap. The INDELs were treated separately from SNPs, since they represent structural rather than substitutional variation.

The INDEL analysis showed that the sequences presented were small number of insertions and deletions detected across the dataset. The reference sequence itself, along with MF590181.1 and MF590175.1, contained no INDELs. A subset of sequences, including MF590180.1 and MF590174.1, displayed three insertions at the beginning of the alignment (positions 1–3). In contrast, the rest of the sequences showed deletions clustered at the end of the alignment, mostly around positions 238–241. These deletions were observed in MF590179.1, MF590178.1, MF590177.1, MF590176.1, MF590174.1, and MF590173.1, with MF590173.1 showing the largest number of deletions in this region, with 4 deletions.

In [26]:
```
## Logic steps:

#1) Start an empty list to store INDELs.
#2) Use "count" as the position counter.
#3) Loop through every position in the sequences.
#4) If reference has a base and sequence has a gap → deletion.
#5) If reference has a gap and sequence has a base → insertion.
#6) Save each INDEL as (position, type, base).
```

```
#7) Move to the next position until the end.
#8) Return the list of INDELs.
def list_indels(ref, seq):
    '''
    Finds INDELs between the reference sequence and another sequence.
    '''

    indels = []
    count = 0

    while count < len(ref):
        r = ref[count]
        s = seq[count]

        if r != s:
            if r != "-" and s == "-":
                indels.append((count + 1, "deletion", r))
            elif r == "-" and s != "-":
                indels.append((count + 1, "insertion", s))

        count = count + 1

    return indels

# LOGIC STEPS
# 1) choose the first sequence as the reference
# 2) convert the reference to a string
# 3) get how many sequences exist (n = number of sequences)
# 4) loop through each sequence one by one
# 5) convert the current sequence to a string
# 6) compare it to the reference using list_indels()
# 7) print the number of INDELs and their details
# 8) continue until all sequences are processed

def identify_indels(records):
    '''
    Identifies INDELs in all sequences relative to the first sequence,
    treating the first sequence as the reference.
    '''

    ref = str(records[0].seq)
    n = len(records)

    print("INDELs relative to reference:", records[0].id, "\n")

    count = 0
    while count < n:
        seq = str(records[count].seq)
        indels = list_indels(ref, seq)

        print(f"{records[count].id} ({len(indels)} INDELs)")
        for pos, indel_type, base in indels:
            print(f"  Position {pos}: {indel_type} ({base})")
        print()

        count = count + 1
```

In [44]:
```
identify_indels(records)
```

```
INDELs relative to reference: lcl|MF590182.1_cds_AWA45334.1_1

lcl|MF590182.1_cds_AWA45334.1_1 (0 INDELs)

lcl|MF590181.1_cds_AWA45333.1_1 (0 INDELs)

lcl|MF590180.1_cds_AWA45332.1_1 (3 INDELs)
  Position 1: insertion (g)
  Position 2: insertion (a)
  Position 3: insertion (g)

lcl|MF590179.1_cds_AWA45331.1_1 (2 INDELs)
  Position 240: deletion (c)
  Position 241: deletion (a)

lcl|MF590178.1_cds_AWA45330.1_1 (1 INDELs)
  Position 241: deletion (a)

lcl|MF590177.1_cds_AWA45329.1_1 (2 INDELs)
  Position 240: deletion (c)
  Position 241: deletion (a)

lcl|MF590176.1_cds_AWA45328.1_1 (3 INDELs)
  Position 239: deletion (c)
  Position 240: deletion (c)
  Position 241: deletion (a)

lcl|MF590175.1_cds_AWA45327.1_1 (0 INDELs)

lcl|MF590174.1_cds_AWA45326.1_1 (5 INDELs)
  Position 1: insertion (g)
  Position 2: insertion (a)
  Position 3: insertion (g)
  Position 240: deletion (c)
  Position 241: deletion (a)

lcl|MF590173.1_cds_AWA45325.1_1 (4 INDELs)
  Position 238: deletion (c)
  Position 239: deletion (c)
  Position 240: deletion (c)
  Position 241: deletion (a)
```

ANALISYS OF DATASET 2 #(BRCA1 PARTIAL GENE ISOLATES) AND RESULTS

The same analysis pipeline was applied to the second FASTA file, which contained the partial BRCA1 gene sequences; all steps were repeated by simply calling the previously defined functions.

In [29]:
```
# loading the file into python  and parsing the file
## logic steps
# call the dunction to parse the file
BCRA1_file_2 = 'C:/Users/user/Downloads/BRCA1_partial_gene_isolates.fasta'
print (parse_file (BCRA1_file_2, 'fasta'))
```

{'lcl|MF945608.1_cds_AXG51130.1_1': 'CTCACTAAAGACAGAATGAATGTAGAAAAGGCTGAATTCTGTAATAAA
AGCAAACAGCCTGGCTTAGCAAGGAGCCAACATAACAGATGGGCTGGAAGTAAGGAAACATGTAATGATAGGCGGACTCCCAGCA
CAGAAAAAAAGGTAGATCTGAATGCTGATCCCCTGTGTGAGAGAAAAGAATGGAATAAGCAGAAACTGCCATGCTCAGAGAATCC
TAGAGATACTGAAGATGTTCCTTGGATAACACTAAATAGCAGCATTCAGAAAG', 'lcl|MF945607.1_cds_AXG5112
9.1_1': 'CTCACTAAAGACAGAATGAATGTAGAAAAGGCTGAATTCTGTAATAAAAGCAAACAGCCTGGCTTAGCAAGGAGCC
AACATAACAGATGGGCTGGAAGTAAGGAAACATGTAATGATAGGCGGACTCCCAGCACAGAAAAAAAGGTAGATCTGAATGCTGA
TCCCCTGTGTGAGAGAAAAGAATGGAATAAGCAGAAACTGCCATGCTCAGAGAATCCTAGAGATACTGAAGATGTTCCTTGGATA
ACACTAAATAGCAGCATTCAGAAAG', 'lcl|MF945606.1_cds_AXG51128.1_1': 'CTCACTAAAGACAGAATGAAT
GTAGAAAAGGCTGAATTCTGTAATAAAAGCAAACAGCCTGGCTTAGCAAGGAGCCAACATAACAGATGGGCTGGAAGTAAGGAAA
CATGTAATGATAGGCGGACTCCCAGCACAGAAAAAAAGGTAGATCTGAATGCTGATCCCCTGTGTGAGAGAAAAGAATGGAATAA
GCGGAAACTGCCATGCTCAGAGAATCCTAGAGATACTGAAGATGTTCCTTGGATAACACTAAATAGCAGCATTCAGAAAG', 'l
cl|MF945605.1_cds_AXG51127.1_1': 'AGAATGAATGTAGAAAAGGCTGAATTCTGTAATAAAAGCAAACAGCCTGGC
TTAGCAAGGAGCCAACATAACAGATGGGCTGGAAGTAAGGAAACATGTAATGATAGGCGGACTCCCAGCACAGAAAAAAAGGTAG
ATCTGAATGCTGATCCCCTGTGTGAGAGAAAAGAATGGAATAAGCAGAAACTGCCATGCTCAGAGAATCCTAGAGATACTGAAGA
TGTTCCTTGGATAACACTAAATAGCAGCATTCAGAAAG', 'lcl|MF945604.1_cds_AXG51126.1_1': 'AGAATGAA
TGTAGAAAAGGCTGAATTCTGTAATAAAAGCAAACAGCCTGGCTTAGCAAGGAGCCAACATAACAGATGGGCTGGAAGTAAGGAA
ACATGTAATGATAGGCGGACTCCCAGCACAGAAAAAAAGGTAGATCTGAATGCTGATCCCCTGTGTGAGAGAAAAGAATGGAATA
AGCAGAAACTGCCATGCTCAGAGAATCCTAGAGATACTGAAGATGTTCCTTGGATAACACTAAATAGCAGCATTCAGAAAG',
'lcl|MF945603.1_cds_AXG51125.1_1': 'CTCACTAAAGACAGAATGAATGTAGAAAAGGCTGAATTCTGTAATAAAA
GCAAACAGCCTGGCTTAGCAAGGAGCCAACATAACAGATGGGCTGGAAGTAAGGAAACATGTAATGATAGGCGGACTCCCAGCAC
AGAAAAAAAGGTAGATCTGAATGCTGATCCCCTGTGTGAGAGAAAAGAATGGAATAAGCAGAAACTGCCATGCTCAGAGAATCCT
AGAGATAC', 'lcl|MF945602.1_cds_AXG51124.1_1': 'CTCACTAAAGACAGAATGAATGTAGAAAAGGCTGAATT
CTGTAATAAAAGCAAACAGCCTGGCTTAGCAAGGAGCCAACATAACAGATGGGCTGGAAGTAAGGAAACATGTAATGATAGGCGG
ACTCCCAGCACAGAAAAAAAGGTAGATCTGAATGCTGATCCCCTGTGTGAGAGAAAAGAATGGAATAAGCAGAAACTGCCATGCT
CAGAGAATCCTAGAGATACTGAAGATGTTCCTTGGATAACACTAAATAGCAGCATTCAGAAAG', 'lcl|MF945601.1_cds
_AXG51123.1_1': 'AGCAAACAGCCTGGCTTAGCAAGGAGCCAACATAACAGATGGGCTGGAAGTAAGGAAACATGTAATGA
TAGGCGGACTCCCAGCACAGAAAAAAAGGTAGATCTGAATGCTGATCCCCTGTGTGAGAGAAAAGAATGGAATAAGCAGAAACTG
CCATGCTCAGAGAATCCTAGAGATACTGAAGATGTTCCTTGGATAACACTAAATAGCAGCATTCAGAAAG', 'lcl|KT12006
1.1_cds_ALF35866.1_1': 'ATGGATTTATCTGCTCTTCGCGTTGAAGAAGTACAAAATGTCATTAATGCTATGCAGAAAA
TCTTAGAGTGTCCCATCTC'}

1. preparing the sequences for alignment

In [ ]:
```
data2 = parse_file(BCRA1_file_2, 'fasta')
records_file_2 = seq_records(data2)
temporary_file_2 = write_fasta(records_file_2, "temp2.fasta")
records2 = list(SeqIO.parse("aligned2.fasta", "fasta"))
aligned_file1 = align_with_mafft("C:\\Users\\user\\Downloads\\temp2.fasta")
```

1. Similarity score between sequences

The similarity analysis of the partial BRCA1 gene isolates revealed that four sequences had high pairwise similarity values ranging from 92% to 97%. The sequences MF945608.1, MF945607.1, MF945606.1, and MF945602.1 consistently formed the most conserved subgroup, each sharing 97% similarity, indicating that these isolates represent closely related variants of the same gene fragment. Two additional sequences (MF945605.1 and MF945604.1) showed moderately similar values of 92–93% to the core group, suggesting they belong to the same lineage but contain slightly more divergence. In contrast, MF945603.1 and MF945601.1 presented more reduced similarity values (64–81%), making them more distant from the others. The sequence KT120061.1 showed only 15% similarity to all other isolates, suggesting that it may not belong to the same group as the other sequences. This is explained by the phylogenetic tree bellow (figure 2).

In [36]:
```
compare_all(records2)
```

```
lcl|MF945608.1_cds_AXG51130.1_1 vs lcl|MF945607.1_cds_AXG51129.1_1 = 97 %
lcl|MF945608.1_cds_AXG51130.1_1 vs lcl|MF945606.1_cds_AXG51128.1_1 = 97 %
lcl|MF945608.1_cds_AXG51130.1_1 vs lcl|MF945605.1_cds_AXG51127.1_1 = 93 %
lcl|MF945608.1_cds_AXG51130.1_1 vs lcl|MF945604.1_cds_AXG51126.1_1 = 93 %
lcl|MF945608.1_cds_AXG51130.1_1 vs lcl|MF945603.1_cds_AXG51125.1_1 = 81 %
lcl|MF945608.1_cds_AXG51130.1_1 vs lcl|MF945602.1_cds_AXG51124.1_1 = 97 %
lcl|MF945608.1_cds_AXG51130.1_1 vs lcl|MF945601.1_cds_AXG51123.1_1 = 80 %
lcl|MF945608.1_cds_AXG51130.1_1 vs lcl|KT120061.1_cds_ALF35866.1_1 = 15 %
lcl|MF945607.1_cds_AXG51129.1_1 vs lcl|MF945606.1_cds_AXG51128.1_1 = 97 %
lcl|MF945607.1_cds_AXG51129.1_1 vs lcl|MF945605.1_cds_AXG51127.1_1 = 93 %
lcl|MF945607.1_cds_AXG51129.1_1 vs lcl|MF945604.1_cds_AXG51126.1_1 = 93 %
lcl|MF945607.1_cds_AXG51129.1_1 vs lcl|MF945603.1_cds_AXG51125.1_1 = 81 %
lcl|MF945607.1_cds_AXG51129.1_1 vs lcl|MF945602.1_cds_AXG51124.1_1 = 97 %
lcl|MF945607.1_cds_AXG51129.1_1 vs lcl|MF945601.1_cds_AXG51123.1_1 = 80 %
lcl|MF945607.1_cds_AXG51129.1_1 vs lcl|KT120061.1_cds_ALF35866.1_1 = 15 %
lcl|MF945606.1_cds_AXG51128.1_1 vs lcl|MF945605.1_cds_AXG51127.1_1 = 92 %
lcl|MF945606.1_cds_AXG51128.1_1 vs lcl|MF945604.1_cds_AXG51126.1_1 = 92 %
lcl|MF945606.1_cds_AXG51128.1_1 vs lcl|MF945603.1_cds_AXG51125.1_1 = 81 %
lcl|MF945606.1_cds_AXG51128.1_1 vs lcl|MF945602.1_cds_AXG51124.1_1 = 97 %
lcl|MF945606.1_cds_AXG51128.1_1 vs lcl|MF945601.1_cds_AXG51123.1_1 = 79 %
lcl|MF945606.1_cds_AXG51128.1_1 vs lcl|KT120061.1_cds_ALF35866.1_1 = 15 %
lcl|MF945605.1_cds_AXG51127.1_1 vs lcl|MF945604.1_cds_AXG51126.1_1 = 93 %
lcl|MF945605.1_cds_AXG51127.1_1 vs lcl|MF945603.1_cds_AXG51125.1_1 = 77 %
lcl|MF945605.1_cds_AXG51127.1_1 vs lcl|MF945602.1_cds_AXG51124.1_1 = 93 %
lcl|MF945605.1_cds_AXG51127.1_1 vs lcl|MF945601.1_cds_AXG51123.1_1 = 80 %
lcl|MF945605.1_cds_AXG51127.1_1 vs lcl|KT120061.1_cds_ALF35866.1_1 = 15 %
lcl|MF945604.1_cds_AXG51126.1_1 vs lcl|MF945603.1_cds_AXG51125.1_1 = 77 %
lcl|MF945604.1_cds_AXG51126.1_1 vs lcl|MF945602.1_cds_AXG51124.1_1 = 93 %
lcl|MF945604.1_cds_AXG51126.1_1 vs lcl|MF945601.1_cds_AXG51123.1_1 = 80 %
lcl|MF945604.1_cds_AXG51126.1_1 vs lcl|KT120061.1_cds_ALF35866.1_1 = 15 %
lcl|MF945603.1_cds_AXG51125.1_1 vs lcl|MF945602.1_cds_AXG51124.1_1 = 81 %
lcl|MF945603.1_cds_AXG51125.1_1 vs lcl|MF945601.1_cds_AXG51123.1_1 = 64 %
lcl|MF945603.1_cds_AXG51125.1_1 vs lcl|KT120061.1_cds_ALF35866.1_1 = 15 %
lcl|MF945602.1_cds_AXG51124.1_1 vs lcl|MF945601.1_cds_AXG51123.1_1 = 80 %
lcl|MF945602.1_cds_AXG51124.1_1 vs lcl|KT120061.1_cds_ALF35866.1_1 = 15 %
lcl|MF945601.1_cds_AXG51123.1_1 vs lcl|KT120061.1_cds_ALF35866.1_1 = 15 %
```

Phylogenetic tree

The partial BRCA1 gene fragments. Unlike the full-length mRNA isolates, the partial sequences form multiple distinct branches, reflecting substantial variation in sequence integrity. A small subgroup of isolates clusters tightly together, indicating limited divergence and preservation of a shared core fragment. Moderately truncated sequences form intermediate branches, consistent with their increased structural disruption. The most fragmented isolate forms a long, independent branch, highlighting its extreme divergence from the rest of the dataset (Figure 2).

In [2]:

```python
import numpy as np
from scipy.cluster.hierarchy import dendrogram, linkage
import matplotlib.pyplot as plt

# the names of your sequences
names = [
    "MF945608.1", "MF945607.1", "MF945606.1", "MF945605.1",
    "MF945604.1", "MF945603.1", "MF945602.1", "MF945601.1",
    "KT120061.1"
]

# my similarity matrix (in %)
similarity = np.array([
    [100, 97, 97, 93, 93, 81, 97, 80, 15],
    [97, 100, 97, 93, 93, 81, 97, 80, 15],
    [97, 97, 100, 92, 92, 81, 97, 79, 15],
    [93, 93, 92, 100, 93, 77, 93, 80, 15],
    [93, 93, 92, 93, 100, 77, 93, 80, 15],
```

```
    [81, 81, 81, 77, 77, 100, 81, 64, 15],
    [97, 97, 97, 93, 93, 81, 100, 80, 15],
    [80, 80, 79, 80, 80, 64, 80, 100, 15],
    [15, 15, 15, 15, 15, 15, 15, 15, 100]
])

# Convert similarity (%) to distance (0 = identical, 1 = completely different)
distance = 1 - (similarity / 100)

# Convert full matrix to condensed form required by linkage()
from scipy.spatial.distance import squareform

condensed_dist = squareform(distance)

# Build the tree using UPGMA (method='average')
Z = linkage(condensed_dist, method='average')

plt.figure(figsize=(10, 6))
dendrogram(Z, labels=names, leaf_rotation=90)
plt.title("Phylogenetic Tree (UPGMA) – Partial BRCA1 Isolates")
plt.ylabel("Distance")
plt.tight_layout()
plt.show()
```

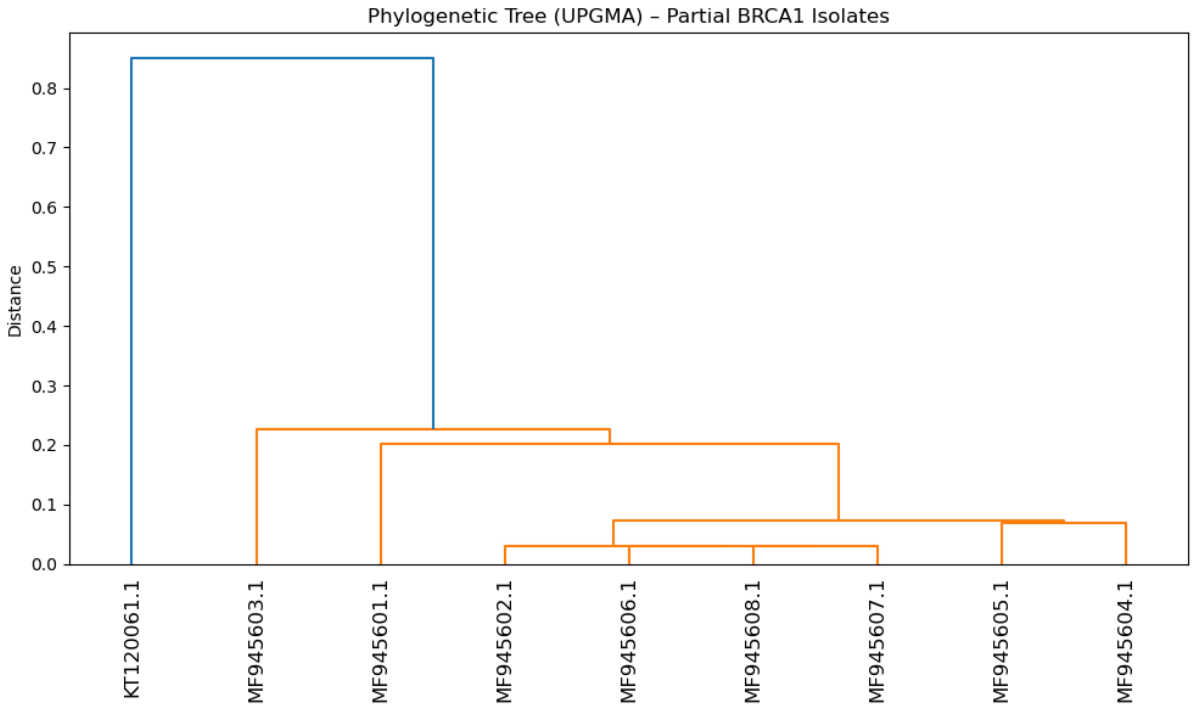


Figure 2. Phylogenetic tree of BRCA1 partiar gene isolates (File 2)

2. Differences between the sequences

The difference analysis justifies the previous similarities score in the partial gene sequences when compared to each other; the differences come mainly from long gaps at the beginning or end of the sequences, and some base changes. The three sequences MF945608.1, MF945607.1, and MF945606.1 are almost identical, with only one small base change around position 194, where one sequence has an *a*, and the others have a *g*. The sequences MF945605.1 and MF945604.1 both show a block of gaps at positions 1–12, meaning they are missing the first twelve bases (ctcactaaagac) compared to the others. The sequence MF945603.1 is missing a large section at the end, with continuous gaps from positions 235 to 278, showing that this isolate lacks the final part of the gene fragment. MF945601.1 has even more gaps, starting from the very beginning and continuing through many positions, making it much shorter than the rest. The most different sequence, KT120061.1, contains gaps across almost

the entire alignment and also shows many mismatched bases, confirming that it does not belong to the same group.

In [37]:
```
show_all_differences(records2)
```

```
Comparing: lcl|MF945608.1_cds_AXG51130.1_1 vs lcl|MF945607.1_cds_AXG51129.1_1

Comparing: lcl|MF945608.1_cds_AXG51130.1_1 vs lcl|MF945606.1_cds_AXG51128.1_1
Position 194 : a vs g

Comparing: lcl|MF945608.1_cds_AXG51130.1_1 vs lcl|MF945605.1_cds_AXG51127.1_1
Position 1 : c vs -
Position 2 : t vs -
Position 3 : c vs -
Position 4 : a vs -
Position 5 : c vs -
Position 6 : t vs -
Position 7 : a vs -
Position 8 : a vs -
Position 9 : a vs -
Position 10 : g vs -
Position 11 : a vs -
Position 12 : c vs -

Comparing: lcl|MF945608.1_cds_AXG51130.1_1 vs lcl|MF945604.1_cds_AXG51126.1_1
Position 1 : c vs -
Position 2 : t vs -
Position 3 : c vs -
Position 4 : a vs -
Position 5 : c vs -
Position 6 : t vs -
Position 7 : a vs -
Position 8 : a vs -
Position 9 : a vs -
Position 10 : g vs -
Position 11 : a vs -
Position 12 : c vs -

Comparing: lcl|MF945608.1_cds_AXG51130.1_1 vs lcl|MF945603.1_cds_AXG51125.1_1
Position 235 : t vs -
Position 236 : g vs -
Position 237 : a vs -
Position 238 : a vs -
Position 239 : g vs -
Position 240 : a vs -
Position 241 : t vs -
Position 242 : g vs -
Position 243 : t vs -
Position 244 : t vs -
Position 245 : c vs -
Position 246 : c vs -
Position 247 : t vs -
Position 248 : t vs -
Position 249 : g vs -
Position 250 : g vs -
Position 251 : a vs -
Position 252 : t vs -
Position 253 : a vs -
Position 254 : a vs -
Position 255 : c vs -
Position 256 : a vs -
Position 257 : c vs -
Position 258 : t vs -
Position 259 : a vs -
Position 260 : a vs -
Position 261 : a vs -
Position 262 : t vs -
Position 263 : a vs -
Position 264 : g vs -
Position 265 : c vs -
Position 266 : a vs -
Position 267 : g vs -
```

```
Position 268 : c vs -
Position 269 : a vs -
Position 270 : t vs -
Position 271 : t vs -
Position 272 : c vs -
Position 273 : a vs -
Position 274 : g vs -
Position 275 : a vs -
Position 276 : a vs -
Position 277 : a vs -
Position 278 : g vs -

Comparing: lcl|MF945608.1_cds_AXG51130.1_1 vs lcl|MF945602.1_cds_AXG51124.1_1

Comparing: lcl|MF945608.1_cds_AXG51130.1_1 vs lcl|MF945601.1_cds_AXG51123.1_1
Position 1 : c vs -
Position 2 : t vs -
Position 3 : c vs -
Position 4 : a vs -
Position 5 : c vs -
Position 6 : t vs -
Position 7 : a vs -
Position 8 : a vs -
Position 9 : a vs -
Position 10 : g vs -
Position 11 : a vs -
Position 12 : c vs -
Position 13 : a vs -
Position 14 : g vs -
Position 15 : a vs -
Position 16 : a vs -
Position 17 : t vs -
Position 18 : g vs -
Position 19 : a vs -
Position 20 : a vs -
Position 21 : t vs -
Position 22 : g vs -
Position 23 : t vs -
Position 24 : a vs -
Position 25 : g vs -
Position 26 : a vs -
Position 27 : a vs -
Position 28 : a vs -
Position 29 : a vs -
Position 30 : g vs -
Position 31 : g vs -
Position 32 : c vs -
Position 33 : t vs -
Position 34 : g vs -
Position 35 : a vs -
Position 36 : a vs -
Position 37 : t vs -
Position 38 : t vs -
Position 39 : c vs -
Position 40 : t vs -
Position 41 : g vs -
Position 42 : t vs -
Position 43 : a vs -
Position 44 : a vs -
Position 45 : t vs -
Position 46 : a vs -
Position 47 : a vs -
Position 48 : a vs -

Comparing: lcl|MF945608.1_cds_AXG51130.1_1 vs lcl|KT120061.1_cds_ALF35866.1_1
Position 1 : c vs -
Position 2 : t vs -
```

Position 3 : c vs -
Position 4 : a vs -
Position 5 : c vs -
Position 6 : t vs -
Position 7 : a vs -
Position 8 : a vs -
Position 9 : a vs -
Position 10 : g vs -
Position 11 : a vs -
Position 12 : c vs -
Position 13 : a vs -
Position 14 : g vs -
Position 15 : a vs -
Position 16 : a vs -
Position 17 : t vs -
Position 18 : g vs -
Position 19 : a vs -
Position 20 : a vs -
Position 21 : t vs -
Position 22 : g vs -
Position 23 : t vs -
Position 24 : a vs -
Position 25 : g vs -
Position 26 : a vs -
Position 27 : a vs -
Position 28 : a vs -
Position 29 : a vs -
Position 30 : g vs -
Position 31 : g vs -
Position 32 : c vs -
Position 33 : t vs -
Position 34 : g vs -
Position 35 : a vs -
Position 36 : a vs -
Position 37 : t vs -
Position 38 : t vs -
Position 39 : c vs -
Position 40 : t vs -
Position 41 : g vs -
Position 42 : t vs -
Position 43 : a vs -
Position 44 : a vs -
Position 45 : t vs -
Position 46 : a vs -
Position 47 : a vs -
Position 48 : a vs -
Position 49 : a vs -
Position 50 : g vs -
Position 51 : c vs -
Position 52 : a vs -
Position 53 : a vs -
Position 54 : a vs -
Position 55 : c vs -
Position 56 : a vs -
Position 57 : g vs -
Position 58 : c vs -
Position 59 : c vs -
Position 60 : t vs -
Position 61 : g vs -
Position 62 : g vs -
Position 63 : c vs -
Position 64 : t vs -
Position 65 : t vs -
Position 66 : a vs -
Position 67 : g vs -
Position 68 : c vs -
Position 69 : a vs -

Position 70 : a vs -
Position 71 : g vs -
Position 72 : g vs -
Position 73 : a vs -
Position 74 : g vs -
Position 75 : c vs -
Position 76 : c vs -
Position 77 : a vs -
Position 78 : a vs -
Position 79 : c vs -
Position 80 : a vs -
Position 81 : t vs -
Position 82 : a vs -
Position 83 : a vs -
Position 84 : c vs -
Position 85 : a vs -
Position 86 : g vs -
Position 87 : a vs -
Position 88 : t vs -
Position 89 : g vs -
Position 90 : g vs -
Position 91 : g vs -
Position 92 : c vs -
Position 93 : t vs -
Position 94 : g vs -
Position 95 : g vs -
Position 96 : a vs -
Position 97 : a vs -
Position 98 : g vs -
Position 99 : t vs -
Position 100 : a vs -
Position 101 : a vs -
Position 102 : g vs -
Position 103 : g vs -
Position 104 : a vs -
Position 105 : a vs -
Position 106 : a vs -
Position 107 : c vs -
Position 108 : a vs -
Position 109 : t vs -
Position 110 : g vs -
Position 111 : t vs -
Position 112 : a vs -
Position 113 : a vs -
Position 114 : t vs -
Position 115 : g vs -
Position 116 : a vs -
Position 117 : t vs -
Position 118 : a vs -
Position 119 : g vs -
Position 120 : g vs -
Position 121 : c vs -
Position 122 : g vs -
Position 123 : g vs -
Position 124 : a vs -
Position 125 : c vs -
Position 126 : t vs -
Position 127 : c vs -
Position 128 : c vs -
Position 129 : c vs -
Position 130 : a vs -
Position 131 : g vs -
Position 132 : c vs -
Position 133 : a vs -
Position 134 : c vs -
Position 135 : a vs -
Position 136 : g vs -

Position 137 : a vs -
Position 138 : a vs -
Position 139 : a vs -
Position 140 : a vs -
Position 141 : a vs -
Position 142 : a vs -
Position 143 : a vs -
Position 144 : g vs -
Position 145 : g vs a
Position 147 : a vs g
Position 151 : c vs t
Position 153 : g vs a
Position 154 : a vs t
Position 155 : a vs c
Position 160 : g vs c
Position 161 : a vs t
Position 164 : c vs -
Position 165 : c vs -
Position 166 : c vs -
Position 167 : t vs -
Position 168 : g vs -
Position 169 : t vs -
Position 170 : g vs -
Position 171 : t vs -
Position 172 : g vs -
Position 173 : a vs -
Position 174 : g vs -
Position 175 : a vs -
Position 176 : g vs -
Position 177 : a vs -
Position 178 : a vs -
Position 179 : a vs -
Position 180 : a vs -
Position 182 : a vs c
Position 183 : a vs g
Position 185 : g vs t
Position 189 : t vs g
Position 193 : c vs t
Position 195 : g vs c
Position 199 : c vs a
Position 202 : c vs t
Position 205 : - vs t
Position 206 : - vs t
Position 207 : - vs a
Position 208 : - vs a
Position 209 : - vs t
Position 210 : - vs g
Position 211 : - vs c
Position 213 : g vs a
Position 214 : c vs t
Position 215 : t vs g
Position 220 : g vs a
Position 225 : c vs t
Position 231 : a vs t
Position 232 : t vs g
Position 233 : a vs t
Position 235 : t vs c
Position 236 : g vs c
Position 238 : a vs t
Position 239 : g vs c
Position 240 : a vs t
Position 241 : t vs c
Position 242 : g vs -
Position 243 : t vs -
Position 244 : t vs -
Position 245 : c vs -
Position 246 : c vs -

```
Position 247 : t vs -
Position 248 : t vs -
Position 249 : g vs -
Position 250 : g vs -
Position 251 : a vs -
Position 252 : t vs -
Position 253 : a vs -
Position 254 : a vs -
Position 255 : c vs -
Position 256 : a vs -
Position 257 : c vs -
Position 258 : t vs -
Position 259 : a vs -
Position 260 : a vs -
Position 261 : a vs -
Position 262 : t vs -
Position 263 : a vs -
Position 264 : g vs -
Position 265 : c vs -
Position 266 : a vs -
Position 267 : g vs -
Position 268 : c vs -
Position 269 : a vs -
Position 270 : t vs -
Position 271 : t vs -
Position 272 : c vs -
Position 273 : a vs -
Position 274 : g vs -
Position 275 : a vs -
Position 276 : a vs -
Position 277 : a vs -
Position 278 : g vs -

Comparing: lcl|MF945607.1_cds_AXG51129.1_1 vs lcl|MF945606.1_cds_AXG51128.1_1
Position 194 : a vs g

Comparing: lcl|MF945607.1_cds_AXG51129.1_1 vs lcl|MF945605.1_cds_AXG51127.1_1
Position 1 : c vs -
Position 2 : t vs -
Position 3 : c vs -
Position 4 : a vs -
Position 5 : c vs -
Position 6 : t vs -
Position 7 : a vs -
Position 8 : a vs -
Position 9 : a vs -
Position 10 : g vs -
Position 11 : a vs -
Position 12 : c vs -

Comparing: lcl|MF945607.1_cds_AXG51129.1_1 vs lcl|MF945604.1_cds_AXG51126.1_1
Position 1 : c vs -
Position 2 : t vs -
Position 3 : c vs -
Position 4 : a vs -
Position 5 : c vs -
Position 6 : t vs -
Position 7 : a vs -
Position 8 : a vs -
Position 9 : a vs -
Position 10 : g vs -
Position 11 : a vs -
Position 12 : c vs -

Comparing: lcl|MF945607.1_cds_AXG51129.1_1 vs lcl|MF945603.1_cds_AXG51125.1_1
Position 235 : t vs -
Position 236 : g vs -
```

```
Position 237 : a vs -
Position 238 : a vs -
Position 239 : g vs -
Position 240 : a vs -
Position 241 : t vs -
Position 242 : g vs -
Position 243 : t vs -
Position 244 : t vs -
Position 245 : c vs -
Position 246 : c vs -
Position 247 : t vs -
Position 248 : t vs -
Position 249 : g vs -
Position 250 : g vs -
Position 251 : a vs -
Position 252 : t vs -
Position 253 : a vs -
Position 254 : a vs -
Position 255 : c vs -
Position 256 : a vs -
Position 257 : c vs -
Position 258 : t vs -
Position 259 : a vs -
Position 260 : a vs -
Position 261 : a vs -
Position 262 : t vs -
Position 263 : a vs -
Position 264 : g vs -
Position 265 : c vs -
Position 266 : a vs -
Position 267 : g vs -
Position 268 : c vs -
Position 269 : a vs -
Position 270 : t vs -
Position 271 : t vs -
Position 272 : c vs -
Position 273 : a vs -
Position 274 : g vs -
Position 275 : a vs -
Position 276 : a vs -
Position 277 : a vs -
Position 278 : g vs -

Comparing: lcl|MF945607.1_cds_AXG51129.1_1 vs lcl|MF945602.1_cds_AXG51124.1_1

Comparing: lcl|MF945607.1_cds_AXG51129.1_1 vs lcl|MF945601.1_cds_AXG51123.1_1
Position 1 : c vs -
Position 2 : t vs -
Position 3 : c vs -
Position 4 : a vs -
Position 5 : c vs -
Position 6 : t vs -
Position 7 : a vs -
Position 8 : a vs -
Position 9 : a vs -
Position 10 : g vs -
Position 11 : a vs -
Position 12 : c vs -
Position 13 : a vs -
Position 14 : g vs -
Position 15 : a vs -
Position 16 : a vs -
Position 17 : t vs -
Position 18 : g vs -
Position 19 : a vs -
Position 20 : a vs -
Position 21 : t vs -
```

```
Position 22 : g vs -
Position 23 : t vs -
Position 24 : a vs -
Position 25 : g vs -
Position 26 : a vs -
Position 27 : a vs -
Position 28 : a vs -
Position 29 : a vs -
Position 30 : g vs -
Position 31 : g vs -
Position 32 : c vs -
Position 33 : t vs -
Position 34 : g vs -
Position 35 : a vs -
Position 36 : a vs -
Position 37 : t vs -
Position 38 : t vs -
Position 39 : c vs -
Position 40 : t vs -
Position 41 : g vs -
Position 42 : t vs -
Position 43 : a vs -
Position 44 : a vs -
Position 45 : t vs -
Position 46 : a vs -
Position 47 : a vs -
Position 48 : a vs -

Comparing: lcl|MF945607.1_cds_AXG51129.1_1 vs lcl|KT120061.1_cds_ALF35866.1_1
Position 1 : c vs -
Position 2 : t vs -
Position 3 : c vs -
Position 4 : a vs -
Position 5 : c vs -
Position 6 : t vs -
Position 7 : a vs -
Position 8 : a vs -
Position 9 : a vs -
Position 10 : g vs -
Position 11 : a vs -
Position 12 : c vs -
Position 13 : a vs -
Position 14 : g vs -
Position 15 : a vs -
Position 16 : a vs -
Position 17 : t vs -
Position 18 : g vs -
Position 19 : a vs -
Position 20 : a vs -
Position 21 : t vs -
Position 22 : g vs -
Position 23 : t vs -
Position 24 : a vs -
Position 25 : g vs -
Position 26 : a vs -
Position 27 : a vs -
Position 28 : a vs -
Position 29 : a vs -
Position 30 : g vs -
Position 31 : g vs -
Position 32 : c vs -
Position 33 : t vs -
Position 34 : g vs -
Position 35 : a vs -
Position 36 : a vs -
Position 37 : t vs -
Position 38 : t vs -
```

Position 39 : c vs -
Position 40 : t vs -
Position 41 : g vs -
Position 42 : t vs -
Position 43 : a vs -
Position 44 : a vs -
Position 45 : t vs -
Position 46 : a vs -
Position 47 : a vs -
Position 48 : a vs -
Position 49 : a vs -
Position 50 : g vs -
Position 51 : c vs -
Position 52 : a vs -
Position 53 : a vs -
Position 54 : a vs -
Position 55 : c vs -
Position 56 : a vs -
Position 57 : g vs -
Position 58 : c vs -
Position 59 : c vs -
Position 60 : t vs -
Position 61 : g vs -
Position 62 : g vs -
Position 63 : c vs -
Position 64 : t vs -
Position 65 : t vs -
Position 66 : a vs -
Position 67 : g vs -
Position 68 : c vs -
Position 69 : a vs -
Position 70 : a vs -
Position 71 : g vs -
Position 72 : g vs -
Position 73 : a vs -
Position 74 : g vs -
Position 75 : c vs -
Position 76 : c vs -
Position 77 : a vs -
Position 78 : a vs -
Position 79 : c vs -
Position 80 : a vs -
Position 81 : t vs -
Position 82 : a vs -
Position 83 : a vs -
Position 84 : c vs -
Position 85 : a vs -
Position 86 : g vs -
Position 87 : a vs -
Position 88 : t vs -
Position 89 : g vs -
Position 90 : g vs -
Position 91 : g vs -
Position 92 : c vs -
Position 93 : t vs -
Position 94 : g vs -
Position 95 : g vs -
Position 96 : a vs -
Position 97 : a vs -
Position 98 : g vs -
Position 99 : t vs -
Position 100 : a vs -
Position 101 : a vs -
Position 102 : g vs -
Position 103 : g vs -
Position 104 : a vs -
Position 105 : a vs -

Position 106 : a vs -
Position 107 : c vs -
Position 108 : a vs -
Position 109 : t vs -
Position 110 : g vs -
Position 111 : t vs -
Position 112 : a vs -
Position 113 : a vs -
Position 114 : t vs -
Position 115 : g vs -
Position 116 : a vs -
Position 117 : t vs -
Position 118 : a vs -
Position 119 : g vs -
Position 120 : g vs -
Position 121 : c vs -
Position 122 : g vs -
Position 123 : g vs -
Position 124 : a vs -
Position 125 : c vs -
Position 126 : t vs -
Position 127 : c vs -
Position 128 : c vs -
Position 129 : c vs -
Position 130 : a vs -
Position 131 : g vs -
Position 132 : c vs -
Position 133 : a vs -
Position 134 : c vs -
Position 135 : a vs -
Position 136 : g vs -
Position 137 : a vs -
Position 138 : a vs -
Position 139 : a vs -
Position 140 : a vs -
Position 141 : a vs -
Position 142 : a vs -
Position 143 : a vs -
Position 144 : g vs -
Position 145 : g vs a
Position 147 : a vs g
Position 151 : c vs t
Position 153 : g vs a
Position 154 : a vs t
Position 155 : a vs c
Position 160 : g vs c
Position 161 : a vs t
Position 164 : c vs -
Position 165 : c vs -
Position 166 : c vs -
Position 167 : t vs -
Position 168 : g vs -
Position 169 : t vs -
Position 170 : g vs -
Position 171 : t vs -
Position 172 : g vs -
Position 173 : a vs -
Position 174 : g vs -
Position 175 : a vs -
Position 176 : g vs -
Position 177 : a vs -
Position 178 : a vs -
Position 179 : a vs -
Position 180 : a vs -
Position 182 : a vs c
Position 183 : a vs g
Position 185 : g vs t

```
Position 189 : t vs g
Position 193 : c vs t
Position 195 : g vs c
Position 199 : c vs a
Position 202 : c vs t
Position 205 : - vs t
Position 206 : - vs t
Position 207 : - vs a
Position 208 : - vs a
Position 209 : - vs t
Position 210 : - vs g
Position 211 : - vs c
Position 213 : g vs a
Position 214 : c vs t
Position 215 : t vs g
Position 220 : g vs a
Position 225 : c vs t
Position 231 : a vs t
Position 232 : t vs g
Position 233 : a vs t
Position 235 : t vs c
Position 236 : g vs c
Position 238 : a vs t
Position 239 : g vs c
Position 240 : a vs t
Position 241 : t vs c
Position 242 : g vs -
Position 243 : t vs -
Position 244 : t vs -
Position 245 : c vs -
Position 246 : c vs -
Position 247 : t vs -
Position 248 : t vs -
Position 249 : g vs -
Position 250 : g vs -
Position 251 : a vs -
Position 252 : t vs -
Position 253 : a vs -
Position 254 : a vs -
Position 255 : c vs -
Position 256 : a vs -
Position 257 : c vs -
Position 258 : t vs -
Position 259 : a vs -
Position 260 : a vs -
Position 261 : a vs -
Position 262 : t vs -
Position 263 : a vs -
Position 264 : g vs -
Position 265 : c vs -
Position 266 : a vs -
Position 267 : g vs -
Position 268 : c vs -
Position 269 : a vs -
Position 270 : t vs -
Position 271 : t vs -
Position 272 : c vs -
Position 273 : a vs -
Position 274 : g vs -
Position 275 : a vs -
Position 276 : a vs -
Position 277 : a vs -
Position 278 : g vs -

Comparing: lcl|MF945606.1_cds_AXG51128.1_1 vs lcl|MF945605.1_cds_AXG51127.1_1
Position 1 : c vs -
Position 2 : t vs -
```

```
Position 3 : c vs -
Position 4 : a vs -
Position 5 : c vs -
Position 6 : t vs -
Position 7 : a vs -
Position 8 : a vs -
Position 9 : a vs -
Position 10 : g vs -
Position 11 : a vs -
Position 12 : c vs -
Position 194 : g vs a

Comparing: lcl|MF945606.1_cds_AXG51128.1_1 vs lcl|MF945604.1_cds_AXG51126.1_1
Position 1 : c vs -
Position 2 : t vs -
Position 3 : c vs -
Position 4 : a vs -
Position 5 : c vs -
Position 6 : t vs -
Position 7 : a vs -
Position 8 : a vs -
Position 9 : a vs -
Position 10 : g vs -
Position 11 : a vs -
Position 12 : c vs -
Position 194 : g vs a

Comparing: lcl|MF945606.1_cds_AXG51128.1_1 vs lcl|MF945603.1_cds_AXG51125.1_1
Position 194 : g vs a
Position 235 : t vs -
Position 236 : g vs -
Position 237 : a vs -
Position 238 : a vs -
Position 239 : g vs -
Position 240 : a vs -
Position 241 : t vs -
Position 242 : g vs -
Position 243 : t vs -
Position 244 : t vs -
Position 245 : c vs -
Position 246 : c vs -
Position 247 : t vs -
Position 248 : t vs -
Position 249 : g vs -
Position 250 : g vs -
Position 251 : a vs -
Position 252 : t vs -
Position 253 : a vs -
Position 254 : a vs -
Position 255 : c vs -
Position 256 : a vs -
Position 257 : c vs -
Position 258 : t vs -
Position 259 : a vs -
Position 260 : a vs -
Position 261 : a vs -
Position 262 : t vs -
Position 263 : a vs -
Position 264 : g vs -
Position 265 : c vs -
Position 266 : a vs -
Position 267 : g vs -
Position 268 : c vs -
Position 269 : a vs -
Position 270 : t vs -
Position 271 : t vs -
Position 272 : c vs -
```

```
Position 273 : a vs -
Position 274 : g vs -
Position 275 : a vs -
Position 276 : a vs -
Position 277 : a vs -
Position 278 : g vs -

Comparing: lcl|MF945606.1_cds_AXG51128.1_1 vs lcl|MF945602.1_cds_AXG51124.1_1
Position 194 : g vs a

Comparing: lcl|MF945606.1_cds_AXG51128.1_1 vs lcl|MF945601.1_cds_AXG51123.1_1
Position 1 : c vs -
Position 2 : t vs -
Position 3 : c vs -
Position 4 : a vs -
Position 5 : c vs -
Position 6 : t vs -
Position 7 : a vs -
Position 8 : a vs -
Position 9 : a vs -
Position 10 : g vs -
Position 11 : a vs -
Position 12 : c vs -
Position 13 : a vs -
Position 14 : g vs -
Position 15 : a vs -
Position 16 : a vs -
Position 17 : t vs -
Position 18 : g vs -
Position 19 : a vs -
Position 20 : a vs -
Position 21 : t vs -
Position 22 : g vs -
Position 23 : t vs -
Position 24 : a vs -
Position 25 : g vs -
Position 26 : a vs -
Position 27 : a vs -
Position 28 : a vs -
Position 29 : a vs -
Position 30 : g vs -
Position 31 : g vs -
Position 32 : c vs -
Position 33 : t vs -
Position 34 : g vs -
Position 35 : a vs -
Position 36 : a vs -
Position 37 : t vs -
Position 38 : t vs -
Position 39 : c vs -
Position 40 : t vs -
Position 41 : g vs -
Position 42 : t vs -
Position 43 : a vs -
Position 44 : a vs -
Position 45 : t vs -
Position 46 : a vs -
Position 47 : a vs -
Position 48 : a vs -
Position 194 : g vs a

Comparing: lcl|MF945606.1_cds_AXG51128.1_1 vs lcl|KT120061.1_cds_ALF35866.1_1
Position 1 : c vs -
Position 2 : t vs -
Position 3 : c vs -
Position 4 : a vs -
Position 5 : c vs -
```

Position 6 : t vs -
Position 7 : a vs -
Position 8 : a vs -
Position 9 : a vs -
Position 10 : g vs -
Position 11 : a vs -
Position 12 : c vs -
Position 13 : a vs -
Position 14 : g vs -
Position 15 : a vs -
Position 16 : a vs -
Position 17 : t vs -
Position 18 : g vs -
Position 19 : a vs -
Position 20 : a vs -
Position 21 : t vs -
Position 22 : g vs -
Position 23 : t vs -
Position 24 : a vs -
Position 25 : g vs -
Position 26 : a vs -
Position 27 : a vs -
Position 28 : a vs -
Position 29 : a vs -
Position 30 : g vs -
Position 31 : g vs -
Position 32 : c vs -
Position 33 : t vs -
Position 34 : g vs -
Position 35 : a vs -
Position 36 : a vs -
Position 37 : t vs -
Position 38 : t vs -
Position 39 : c vs -
Position 40 : t vs -
Position 41 : g vs -
Position 42 : t vs -
Position 43 : a vs -
Position 44 : a vs -
Position 45 : t vs -
Position 46 : a vs -
Position 47 : a vs -
Position 48 : a vs -
Position 49 : a vs -
Position 50 : g vs -
Position 51 : c vs -
Position 52 : a vs -
Position 53 : a vs -
Position 54 : a vs -
Position 55 : c vs -
Position 56 : a vs -
Position 57 : g vs -
Position 58 : c vs -
Position 59 : c vs -
Position 60 : t vs -
Position 61 : g vs -
Position 62 : g vs -
Position 63 : c vs -
Position 64 : t vs -
Position 65 : t vs -
Position 66 : a vs -
Position 67 : g vs -
Position 68 : c vs -
Position 69 : a vs -
Position 70 : a vs -
Position 71 : g vs -
Position 72 : g vs -

Position 73 : a vs -
Position 74 : g vs -
Position 75 : c vs -
Position 76 : c vs -
Position 77 : a vs -
Position 78 : a vs -
Position 79 : c vs -
Position 80 : a vs -
Position 81 : t vs -
Position 82 : a vs -
Position 83 : a vs -
Position 84 : c vs -
Position 85 : a vs -
Position 86 : g vs -
Position 87 : a vs -
Position 88 : t vs -
Position 89 : g vs -
Position 90 : g vs -
Position 91 : g vs -
Position 92 : c vs -
Position 93 : t vs -
Position 94 : g vs -
Position 95 : g vs -
Position 96 : a vs -
Position 97 : a vs -
Position 98 : g vs -
Position 99 : t vs -
Position 100 : a vs -
Position 101 : a vs -
Position 102 : g vs -
Position 103 : g vs -
Position 104 : a vs -
Position 105 : a vs -
Position 106 : a vs -
Position 107 : c vs -
Position 108 : a vs -
Position 109 : t vs -
Position 110 : g vs -
Position 111 : t vs -
Position 112 : a vs -
Position 113 : a vs -
Position 114 : t vs -
Position 115 : g vs -
Position 116 : a vs -
Position 117 : t vs -
Position 118 : a vs -
Position 119 : g vs -
Position 120 : g vs -
Position 121 : c vs -
Position 122 : g vs -
Position 123 : g vs -
Position 124 : a vs -
Position 125 : c vs -
Position 126 : t vs -
Position 127 : c vs -
Position 128 : c vs -
Position 129 : c vs -
Position 130 : a vs -
Position 131 : g vs -
Position 132 : c vs -
Position 133 : a vs -
Position 134 : c vs -
Position 135 : a vs -
Position 136 : g vs -
Position 137 : a vs -
Position 138 : a vs -
Position 139 : a vs -

Position 140 : a vs -
Position 141 : a vs -
Position 142 : a vs -
Position 143 : a vs -
Position 144 : g vs -
Position 145 : g vs a
Position 147 : a vs g
Position 151 : c vs t
Position 153 : g vs a
Position 154 : a vs t
Position 155 : a vs c
Position 160 : g vs c
Position 161 : a vs t
Position 164 : c vs -
Position 165 : c vs -
Position 166 : c vs -
Position 167 : t vs -
Position 168 : g vs -
Position 169 : t vs -
Position 170 : g vs -
Position 171 : t vs -
Position 172 : g vs -
Position 173 : a vs -
Position 174 : g vs -
Position 175 : a vs -
Position 176 : g vs -
Position 177 : a vs -
Position 178 : a vs -
Position 179 : a vs -
Position 180 : a vs -
Position 182 : a vs c
Position 183 : a vs g
Position 185 : g vs t
Position 189 : t vs g
Position 193 : c vs t
Position 194 : g vs a
Position 195 : g vs c
Position 199 : c vs a
Position 202 : c vs t
Position 205 : - vs t
Position 206 : - vs t
Position 207 : - vs a
Position 208 : - vs a
Position 209 : - vs t
Position 210 : - vs g
Position 211 : - vs c
Position 213 : g vs a
Position 214 : c vs t
Position 215 : t vs g
Position 220 : g vs a
Position 225 : c vs t
Position 231 : a vs t
Position 232 : t vs g
Position 233 : a vs t
Position 235 : t vs c
Position 236 : g vs c
Position 238 : a vs t
Position 239 : g vs c
Position 240 : a vs t
Position 241 : t vs c
Position 242 : g vs -
Position 243 : t vs -
Position 244 : t vs -
Position 245 : c vs -
Position 246 : c vs -
Position 247 : t vs -
Position 248 : t vs -

```
Position 249 : g vs -
Position 250 : g vs -
Position 251 : a vs -
Position 252 : t vs -
Position 253 : a vs -
Position 254 : a vs -
Position 255 : c vs -
Position 256 : a vs -
Position 257 : c vs -
Position 258 : t vs -
Position 259 : a vs -
Position 260 : a vs -
Position 261 : a vs -
Position 262 : t vs -
Position 263 : a vs -
Position 264 : g vs -
Position 265 : c vs -
Position 266 : a vs -
Position 267 : g vs -
Position 268 : c vs -
Position 269 : a vs -
Position 270 : t vs -
Position 271 : t vs -
Position 272 : c vs -
Position 273 : a vs -
Position 274 : g vs -
Position 275 : a vs -
Position 276 : a vs -
Position 277 : a vs -
Position 278 : g vs -

Comparing: lcl|MF945605.1_cds_AXG51127.1_1 vs lcl|MF945604.1_cds_AXG51126.1_1

Comparing: lcl|MF945605.1_cds_AXG51127.1_1 vs lcl|MF945603.1_cds_AXG51125.1_1
Position 1 : - vs c
Position 2 : - vs t
Position 3 : - vs c
Position 4 : - vs a
Position 5 : - vs c
Position 6 : - vs t
Position 7 : - vs a
Position 8 : - vs a
Position 9 : - vs a
Position 10 : - vs g
Position 11 : - vs a
Position 12 : - vs c
Position 235 : t vs -
Position 236 : g vs -
Position 237 : a vs -
Position 238 : a vs -
Position 239 : g vs -
Position 240 : a vs -
Position 241 : t vs -
Position 242 : g vs -
Position 243 : t vs -
Position 244 : t vs -
Position 245 : c vs -
Position 246 : c vs -
Position 247 : t vs -
Position 248 : t vs -
Position 249 : g vs -
Position 250 : g vs -
Position 251 : a vs -
Position 252 : t vs -
Position 253 : a vs -
Position 254 : a vs -
Position 255 : c vs -
```

```
Position 256 : a vs -
Position 257 : c vs -
Position 258 : t vs -
Position 259 : a vs -
Position 260 : a vs -
Position 261 : a vs -
Position 262 : t vs -
Position 263 : a vs -
Position 264 : g vs -
Position 265 : c vs -
Position 266 : a vs -
Position 267 : g vs -
Position 268 : c vs -
Position 269 : a vs -
Position 270 : t vs -
Position 271 : t vs -
Position 272 : c vs -
Position 273 : a vs -
Position 274 : g vs -
Position 275 : a vs -
Position 276 : a vs -
Position 277 : a vs -
Position 278 : g vs -

Comparing: lcl|MF945605.1_cds_AXG51127.1_1 vs lcl|MF945602.1_cds_AXG51124.1_1
Position 1 : - vs c
Position 2 : - vs t
Position 3 : - vs c
Position 4 : - vs a
Position 5 : - vs c
Position 6 : - vs t
Position 7 : - vs a
Position 8 : - vs a
Position 9 : - vs a
Position 10 : - vs g
Position 11 : - vs a
Position 12 : - vs c

Comparing: lcl|MF945605.1_cds_AXG51127.1_1 vs lcl|MF945601.1_cds_AXG51123.1_1
Position 13 : a vs -
Position 14 : g vs -
Position 15 : a vs -
Position 16 : a vs -
Position 17 : t vs -
Position 18 : g vs -
Position 19 : a vs -
Position 20 : a vs -
Position 21 : t vs -
Position 22 : g vs -
Position 23 : t vs -
Position 24 : a vs -
Position 25 : g vs -
Position 26 : a vs -
Position 27 : a vs -
Position 28 : a vs -
Position 29 : a vs -
Position 30 : g vs -
Position 31 : g vs -
Position 32 : c vs -
Position 33 : t vs -
Position 34 : g vs -
Position 35 : a vs -
Position 36 : a vs -
Position 37 : t vs -
Position 38 : t vs -
Position 39 : c vs -
Position 40 : t vs -
```

```
Position 41 : g vs -
Position 42 : t vs -
Position 43 : a vs -
Position 44 : a vs -
Position 45 : t vs -
Position 46 : a vs -
Position 47 : a vs -
Position 48 : a vs -

Comparing: lcl|MF945605.1_cds_AXG51127.1_1 vs lcl|KT120061.1_cds_ALF35866.1_1
Position 13 : a vs -
Position 14 : g vs -
Position 15 : a vs -
Position 16 : a vs -
Position 17 : t vs -
Position 18 : g vs -
Position 19 : a vs -
Position 20 : a vs -
Position 21 : t vs -
Position 22 : g vs -
Position 23 : t vs -
Position 24 : a vs -
Position 25 : g vs -
Position 26 : a vs -
Position 27 : a vs -
Position 28 : a vs -
Position 29 : a vs -
Position 30 : g vs -
Position 31 : g vs -
Position 32 : c vs -
Position 33 : t vs -
Position 34 : g vs -
Position 35 : a vs -
Position 36 : a vs -
Position 37 : t vs -
Position 38 : t vs -
Position 39 : c vs -
Position 40 : t vs -
Position 41 : g vs -
Position 42 : t vs -
Position 43 : a vs -
Position 44 : a vs -
Position 45 : t vs -
Position 46 : a vs -
Position 47 : a vs -
Position 48 : a vs -
Position 49 : a vs -
Position 50 : g vs -
Position 51 : c vs -
Position 52 : a vs -
Position 53 : a vs -
Position 54 : a vs -
Position 55 : c vs -
Position 56 : a vs -
Position 57 : g vs -
Position 58 : c vs -
Position 59 : c vs -
Position 60 : t vs -
Position 61 : g vs -
Position 62 : g vs -
Position 63 : c vs -
Position 64 : t vs -
Position 65 : t vs -
Position 66 : a vs -
Position 67 : g vs -
Position 68 : c vs -
Position 69 : a vs -
```

Position 70 : a vs -
Position 71 : g vs -
Position 72 : g vs -
Position 73 : a vs -
Position 74 : g vs -
Position 75 : c vs -
Position 76 : c vs -
Position 77 : a vs -
Position 78 : a vs -
Position 79 : c vs -
Position 80 : a vs -
Position 81 : t vs -
Position 82 : a vs -
Position 83 : a vs -
Position 84 : c vs -
Position 85 : a vs -
Position 86 : g vs -
Position 87 : a vs -
Position 88 : t vs -
Position 89 : g vs -
Position 90 : g vs -
Position 91 : g vs -
Position 92 : c vs -
Position 93 : t vs -
Position 94 : g vs -
Position 95 : g vs -
Position 96 : a vs -
Position 97 : a vs -
Position 98 : g vs -
Position 99 : t vs -
Position 100 : a vs -
Position 101 : a vs -
Position 102 : g vs -
Position 103 : g vs -
Position 104 : a vs -
Position 105 : a vs -
Position 106 : a vs -
Position 107 : c vs -
Position 108 : a vs -
Position 109 : t vs -
Position 110 : g vs -
Position 111 : t vs -
Position 112 : a vs -
Position 113 : a vs -
Position 114 : t vs -
Position 115 : g vs -
Position 116 : a vs -
Position 117 : t vs -
Position 118 : a vs -
Position 119 : g vs -
Position 120 : g vs -
Position 121 : c vs -
Position 122 : g vs -
Position 123 : g vs -
Position 124 : a vs -
Position 125 : c vs -
Position 126 : t vs -
Position 127 : c vs -
Position 128 : c vs -
Position 129 : c vs -
Position 130 : a vs -
Position 131 : g vs -
Position 132 : c vs -
Position 133 : a vs -
Position 134 : c vs -
Position 135 : a vs -
Position 136 : g vs -

Position 137 : a vs -
Position 138 : a vs -
Position 139 : a vs -
Position 140 : a vs -
Position 141 : a vs -
Position 142 : a vs -
Position 143 : a vs -
Position 144 : g vs -
Position 145 : g vs a
Position 147 : a vs g
Position 151 : c vs t
Position 153 : g vs a
Position 154 : a vs t
Position 155 : a vs c
Position 160 : g vs c
Position 161 : a vs t
Position 164 : c vs -
Position 165 : c vs -
Position 166 : c vs -
Position 167 : t vs -
Position 168 : g vs -
Position 169 : t vs -
Position 170 : g vs -
Position 171 : t vs -
Position 172 : g vs -
Position 173 : a vs -
Position 174 : g vs -
Position 175 : a vs -
Position 176 : g vs -
Position 177 : a vs -
Position 178 : a vs -
Position 179 : a vs -
Position 180 : a vs -
Position 182 : a vs c
Position 183 : a vs g
Position 185 : g vs t
Position 189 : t vs g
Position 193 : c vs t
Position 195 : g vs c
Position 199 : c vs a
Position 202 : c vs t
Position 205 : - vs t
Position 206 : - vs t
Position 207 : - vs a
Position 208 : - vs a
Position 209 : - vs t
Position 210 : - vs g
Position 211 : - vs c
Position 213 : g vs a
Position 214 : c vs t
Position 215 : t vs g
Position 220 : g vs a
Position 225 : c vs t
Position 231 : a vs t
Position 232 : t vs g
Position 233 : a vs t
Position 235 : t vs c
Position 236 : g vs c
Position 238 : a vs t
Position 239 : g vs c
Position 240 : a vs t
Position 241 : t vs c
Position 242 : g vs -
Position 243 : t vs -
Position 244 : t vs -
Position 245 : c vs -
Position 246 : c vs -

```
Position 247 : t vs -
Position 248 : t vs -
Position 249 : g vs -
Position 250 : g vs -
Position 251 : a vs -
Position 252 : t vs -
Position 253 : a vs -
Position 254 : a vs -
Position 255 : c vs -
Position 256 : a vs -
Position 257 : c vs -
Position 258 : t vs -
Position 259 : a vs -
Position 260 : a vs -
Position 261 : a vs -
Position 262 : t vs -
Position 263 : a vs -
Position 264 : g vs -
Position 265 : c vs -
Position 266 : a vs -
Position 267 : g vs -
Position 268 : c vs -
Position 269 : a vs -
Position 270 : t vs -
Position 271 : t vs -
Position 272 : c vs -
Position 273 : a vs -
Position 274 : g vs -
Position 275 : a vs -
Position 276 : a vs -
Position 277 : a vs -
Position 278 : g vs -

Comparing: lcl|MF945604.1_cds_AXG51126.1_1 vs lcl|MF945603.1_cds_AXG51125.1_1
Position 1 : - vs c
Position 2 : - vs t
Position 3 : - vs c
Position 4 : - vs a
Position 5 : - vs c
Position 6 : - vs t
Position 7 : - vs a
Position 8 : - vs a
Position 9 : - vs a
Position 10 : - vs g
Position 11 : - vs a
Position 12 : - vs c
Position 235 : t vs -
Position 236 : g vs -
Position 237 : a vs -
Position 238 : a vs -
Position 239 : g vs -
Position 240 : a vs -
Position 241 : t vs -
Position 242 : g vs -
Position 243 : t vs -
Position 244 : t vs -
Position 245 : c vs -
Position 246 : c vs -
Position 247 : t vs -
Position 248 : t vs -
Position 249 : g vs -
Position 250 : g vs -
Position 251 : a vs -
Position 252 : t vs -
Position 253 : a vs -
Position 254 : a vs -
Position 255 : c vs -
```

```
Position 256 : a vs -
Position 257 : c vs -
Position 258 : t vs -
Position 259 : a vs -
Position 260 : a vs -
Position 261 : a vs -
Position 262 : t vs -
Position 263 : a vs -
Position 264 : g vs -
Position 265 : c vs -
Position 266 : a vs -
Position 267 : g vs -
Position 268 : c vs -
Position 269 : a vs -
Position 270 : t vs -
Position 271 : t vs -
Position 272 : c vs -
Position 273 : a vs -
Position 274 : g vs -
Position 275 : a vs -
Position 276 : a vs -
Position 277 : a vs -
Position 278 : g vs -

Comparing: lcl|MF945604.1_cds_AXG51126.1_1 vs lcl|MF945602.1_cds_AXG51124.1_1
Position 1 : - vs c
Position 2 : - vs t
Position 3 : - vs c
Position 4 : - vs a
Position 5 : - vs c
Position 6 : - vs t
Position 7 : - vs a
Position 8 : - vs a
Position 9 : - vs a
Position 10 : - vs g
Position 11 : - vs a
Position 12 : - vs c

Comparing: lcl|MF945604.1_cds_AXG51126.1_1 vs lcl|MF945601.1_cds_AXG51123.1_1
Position 13 : a vs -
Position 14 : g vs -
Position 15 : a vs -
Position 16 : a vs -
Position 17 : t vs -
Position 18 : g vs -
Position 19 : a vs -
Position 20 : a vs -
Position 21 : t vs -
Position 22 : g vs -
Position 23 : t vs -
Position 24 : a vs -
Position 25 : g vs -
Position 26 : a vs -
Position 27 : a vs -
Position 28 : a vs -
Position 29 : a vs -
Position 30 : g vs -
Position 31 : g vs -
Position 32 : c vs -
Position 33 : t vs -
Position 34 : g vs -
Position 35 : a vs -
Position 36 : a vs -
Position 37 : t vs -
Position 38 : t vs -
Position 39 : c vs -
Position 40 : t vs -
```

```
Position 41 : g vs -
Position 42 : t vs -
Position 43 : a vs -
Position 44 : a vs -
Position 45 : t vs -
Position 46 : a vs -
Position 47 : a vs -
Position 48 : a vs -

Comparing: lcl|MF945604.1_cds_AXG51126.1_1 vs lcl|KT120061.1_cds_ALF35866.1_1
Position 13 : a vs -
Position 14 : g vs -
Position 15 : a vs -
Position 16 : a vs -
Position 17 : t vs -
Position 18 : g vs -
Position 19 : a vs -
Position 20 : a vs -
Position 21 : t vs -
Position 22 : g vs -
Position 23 : t vs -
Position 24 : a vs -
Position 25 : g vs -
Position 26 : a vs -
Position 27 : a vs -
Position 28 : a vs -
Position 29 : a vs -
Position 30 : g vs -
Position 31 : g vs -
Position 32 : c vs -
Position 33 : t vs -
Position 34 : g vs -
Position 35 : a vs -
Position 36 : a vs -
Position 37 : t vs -
Position 38 : t vs -
Position 39 : c vs -
Position 40 : t vs -
Position 41 : g vs -
Position 42 : t vs -
Position 43 : a vs -
Position 44 : a vs -
Position 45 : t vs -
Position 46 : a vs -
Position 47 : a vs -
Position 48 : a vs -
Position 49 : a vs -
Position 50 : g vs -
Position 51 : c vs -
Position 52 : a vs -
Position 53 : a vs -
Position 54 : a vs -
Position 55 : c vs -
Position 56 : a vs -
Position 57 : g vs -
Position 58 : c vs -
Position 59 : c vs -
Position 60 : t vs -
Position 61 : g vs -
Position 62 : g vs -
Position 63 : c vs -
Position 64 : t vs -
Position 65 : t vs -
Position 66 : a vs -
Position 67 : g vs -
Position 68 : c vs -
Position 69 : a vs -
```

Position 70 : a vs -
Position 71 : g vs -
Position 72 : g vs -
Position 73 : a vs -
Position 74 : g vs -
Position 75 : c vs -
Position 76 : c vs -
Position 77 : a vs -
Position 78 : a vs -
Position 79 : c vs -
Position 80 : a vs -
Position 81 : t vs -
Position 82 : a vs -
Position 83 : a vs -
Position 84 : c vs -
Position 85 : a vs -
Position 86 : g vs -
Position 87 : a vs -
Position 88 : t vs -
Position 89 : g vs -
Position 90 : g vs -
Position 91 : g vs -
Position 92 : c vs -
Position 93 : t vs -
Position 94 : g vs -
Position 95 : g vs -
Position 96 : a vs -
Position 97 : a vs -
Position 98 : g vs -
Position 99 : t vs -
Position 100 : a vs -
Position 101 : a vs -
Position 102 : g vs -
Position 103 : g vs -
Position 104 : a vs -
Position 105 : a vs -
Position 106 : a vs -
Position 107 : c vs -
Position 108 : a vs -
Position 109 : t vs -
Position 110 : g vs -
Position 111 : t vs -
Position 112 : a vs -
Position 113 : a vs -
Position 114 : t vs -
Position 115 : g vs -
Position 116 : a vs -
Position 117 : t vs -
Position 118 : a vs -
Position 119 : g vs -
Position 120 : g vs -
Position 121 : c vs -
Position 122 : g vs -
Position 123 : g vs -
Position 124 : a vs -
Position 125 : c vs -
Position 126 : t vs -
Position 127 : c vs -
Position 128 : c vs -
Position 129 : c vs -
Position 130 : a vs -
Position 131 : g vs -
Position 132 : c vs -
Position 133 : a vs -
Position 134 : c vs -
Position 135 : a vs -
Position 136 : g vs -

Position 137 : a vs -
Position 138 : a vs -
Position 139 : a vs -
Position 140 : a vs -
Position 141 : a vs -
Position 142 : a vs -
Position 143 : a vs -
Position 144 : g vs -
Position 145 : g vs a
Position 147 : a vs g
Position 151 : c vs t
Position 153 : g vs a
Position 154 : a vs t
Position 155 : a vs c
Position 160 : g vs c
Position 161 : a vs t
Position 164 : c vs -
Position 165 : c vs -
Position 166 : c vs -
Position 167 : t vs -
Position 168 : g vs -
Position 169 : t vs -
Position 170 : g vs -
Position 171 : t vs -
Position 172 : g vs -
Position 173 : a vs -
Position 174 : g vs -
Position 175 : a vs -
Position 176 : g vs -
Position 177 : a vs -
Position 178 : a vs -
Position 179 : a vs -
Position 180 : a vs -
Position 182 : a vs c
Position 183 : a vs g
Position 185 : g vs t
Position 189 : t vs g
Position 193 : c vs t
Position 195 : g vs c
Position 199 : c vs a
Position 202 : c vs t
Position 205 : - vs t
Position 206 : - vs t
Position 207 : - vs a
Position 208 : - vs a
Position 209 : - vs t
Position 210 : - vs g
Position 211 : - vs c
Position 213 : g vs a
Position 214 : c vs t
Position 215 : t vs g
Position 220 : g vs a
Position 225 : c vs t
Position 231 : a vs t
Position 232 : t vs g
Position 233 : a vs t
Position 235 : t vs c
Position 236 : g vs c
Position 238 : a vs t
Position 239 : g vs c
Position 240 : a vs t
Position 241 : t vs c
Position 242 : g vs -
Position 243 : t vs -
Position 244 : t vs -
Position 245 : c vs -
Position 246 : c vs -

```
Position 247 : t vs -
Position 248 : t vs -
Position 249 : g vs -
Position 250 : g vs -
Position 251 : a vs -
Position 252 : t vs -
Position 253 : a vs -
Position 254 : a vs -
Position 255 : c vs -
Position 256 : a vs -
Position 257 : c vs -
Position 258 : t vs -
Position 259 : a vs -
Position 260 : a vs -
Position 261 : a vs -
Position 262 : t vs -
Position 263 : a vs -
Position 264 : g vs -
Position 265 : c vs -
Position 266 : a vs -
Position 267 : g vs -
Position 268 : c vs -
Position 269 : a vs -
Position 270 : t vs -
Position 271 : t vs -
Position 272 : c vs -
Position 273 : a vs -
Position 274 : g vs -
Position 275 : a vs -
Position 276 : a vs -
Position 277 : a vs -
Position 278 : g vs -

Comparing: lcl|MF945603.1_cds_AXG51125.1_1 vs lcl|MF945602.1_cds_AXG51124.1_1
Position 235 : - vs t
Position 236 : - vs g
Position 237 : - vs a
Position 238 : - vs a
Position 239 : - vs g
Position 240 : - vs a
Position 241 : - vs t
Position 242 : - vs g
Position 243 : - vs t
Position 244 : - vs t
Position 245 : - vs c
Position 246 : - vs c
Position 247 : - vs t
Position 248 : - vs t
Position 249 : - vs g
Position 250 : - vs g
Position 251 : - vs a
Position 252 : - vs t
Position 253 : - vs a
Position 254 : - vs a
Position 255 : - vs c
Position 256 : - vs a
Position 257 : - vs c
Position 258 : - vs t
Position 259 : - vs a
Position 260 : - vs a
Position 261 : - vs a
Position 262 : - vs t
Position 263 : - vs a
Position 264 : - vs g
Position 265 : - vs c
Position 266 : - vs a
Position 267 : - vs g
```

```
Position 268 : - vs c
Position 269 : - vs a
Position 270 : - vs t
Position 271 : - vs t
Position 272 : - vs c
Position 273 : - vs a
Position 274 : - vs g
Position 275 : - vs a
Position 276 : - vs a
Position 277 : - vs a
Position 278 : - vs g

Comparing: lcl|MF945603.1_cds_AXG51125.1_1 vs lcl|MF945601.1_cds_AXG51123.1_1
Position 1 : c vs -
Position 2 : t vs -
Position 3 : c vs -
Position 4 : a vs -
Position 5 : c vs -
Position 6 : t vs -
Position 7 : a vs -
Position 8 : a vs -
Position 9 : a vs -
Position 10 : g vs -
Position 11 : a vs -
Position 12 : c vs -
Position 13 : a vs -
Position 14 : g vs -
Position 15 : a vs -
Position 16 : a vs -
Position 17 : t vs -
Position 18 : g vs -
Position 19 : a vs -
Position 20 : a vs -
Position 21 : t vs -
Position 22 : g vs -
Position 23 : t vs -
Position 24 : a vs -
Position 25 : g vs -
Position 26 : a vs -
Position 27 : a vs -
Position 28 : a vs -
Position 29 : a vs -
Position 30 : g vs -
Position 31 : g vs -
Position 32 : c vs -
Position 33 : t vs -
Position 34 : g vs -
Position 35 : a vs -
Position 36 : a vs -
Position 37 : t vs -
Position 38 : t vs -
Position 39 : c vs -
Position 40 : t vs -
Position 41 : g vs -
Position 42 : t vs -
Position 43 : a vs -
Position 44 : a vs -
Position 45 : t vs -
Position 46 : a vs -
Position 47 : a vs -
Position 48 : a vs -
Position 235 : - vs t
Position 236 : - vs g
Position 237 : - vs a
Position 238 : - vs a
Position 239 : - vs g
Position 240 : - vs a
```

```
Position 241 : - vs t
Position 242 : - vs g
Position 243 : - vs t
Position 244 : - vs t
Position 245 : - vs c
Position 246 : - vs c
Position 247 : - vs t
Position 248 : - vs t
Position 249 : - vs g
Position 250 : - vs g
Position 251 : - vs a
Position 252 : - vs t
Position 253 : - vs a
Position 254 : - vs a
Position 255 : - vs c
Position 256 : - vs a
Position 257 : - vs c
Position 258 : - vs t
Position 259 : - vs a
Position 260 : - vs a
Position 261 : - vs a
Position 262 : - vs t
Position 263 : - vs a
Position 264 : - vs g
Position 265 : - vs c
Position 266 : - vs a
Position 267 : - vs g
Position 268 : - vs c
Position 269 : - vs a
Position 270 : - vs t
Position 271 : - vs t
Position 272 : - vs c
Position 273 : - vs a
Position 274 : - vs g
Position 275 : - vs a
Position 276 : - vs a
Position 277 : - vs a
Position 278 : - vs g

Comparing: lcl|MF945603.1_cds_AXG51125.1_1 vs lcl|KT120061.1_cds_ALF35866.1_1
Position 1 : c vs -
Position 2 : t vs -
Position 3 : c vs -
Position 4 : a vs -
Position 5 : c vs -
Position 6 : t vs -
Position 7 : a vs -
Position 8 : a vs -
Position 9 : a vs -
Position 10 : g vs -
Position 11 : a vs -
Position 12 : c vs -
Position 13 : a vs -
Position 14 : g vs -
Position 15 : a vs -
Position 16 : a vs -
Position 17 : t vs -
Position 18 : g vs -
Position 19 : a vs -
Position 20 : a vs -
Position 21 : t vs -
Position 22 : g vs -
Position 23 : t vs -
Position 24 : a vs -
Position 25 : g vs -
Position 26 : a vs -
Position 27 : a vs -
```

```
Position 28 : a vs -
Position 29 : a vs -
Position 30 : g vs -
Position 31 : g vs -
Position 32 : c vs -
Position 33 : t vs -
Position 34 : g vs -
Position 35 : a vs -
Position 36 : a vs -
Position 37 : t vs -
Position 38 : t vs -
Position 39 : c vs -
Position 40 : t vs -
Position 41 : g vs -
Position 42 : t vs -
Position 43 : a vs -
Position 44 : a vs -
Position 45 : t vs -
Position 46 : a vs -
Position 47 : a vs -
Position 48 : a vs -
Position 49 : a vs -
Position 50 : g vs -
Position 51 : c vs -
Position 52 : a vs -
Position 53 : a vs -
Position 54 : a vs -
Position 55 : c vs -
Position 56 : a vs -
Position 57 : g vs -
Position 58 : c vs -
Position 59 : c vs -
Position 60 : t vs -
Position 61 : g vs -
Position 62 : g vs -
Position 63 : c vs -
Position 64 : t vs -
Position 65 : t vs -
Position 66 : a vs -
Position 67 : g vs -
Position 68 : c vs -
Position 69 : a vs -
Position 70 : a vs -
Position 71 : g vs -
Position 72 : g vs -
Position 73 : a vs -
Position 74 : g vs -
Position 75 : c vs -
Position 76 : c vs -
Position 77 : a vs -
Position 78 : a vs -
Position 79 : c vs -
Position 80 : a vs -
Position 81 : t vs -
Position 82 : a vs -
Position 83 : a vs -
Position 84 : c vs -
Position 85 : a vs -
Position 86 : g vs -
Position 87 : a vs -
Position 88 : t vs -
Position 89 : g vs -
Position 90 : g vs -
Position 91 : g vs -
Position 92 : c vs -
Position 93 : t vs -
Position 94 : g vs -
```

```
Position 95 : g vs -
Position 96 : a vs -
Position 97 : a vs -
Position 98 : g vs -
Position 99 : t vs -
Position 100 : a vs -
Position 101 : a vs -
Position 102 : g vs -
Position 103 : g vs -
Position 104 : a vs -
Position 105 : a vs -
Position 106 : a vs -
Position 107 : c vs -
Position 108 : a vs -
Position 109 : t vs -
Position 110 : g vs -
Position 111 : t vs -
Position 112 : a vs -
Position 113 : a vs -
Position 114 : t vs -
Position 115 : g vs -
Position 116 : a vs -
Position 117 : t vs -
Position 118 : a vs -
Position 119 : g vs -
Position 120 : g vs -
Position 121 : c vs -
Position 122 : g vs -
Position 123 : g vs -
Position 124 : a vs -
Position 125 : c vs -
Position 126 : t vs -
Position 127 : c vs -
Position 128 : c vs -
Position 129 : c vs -
Position 130 : a vs -
Position 131 : g vs -
Position 132 : c vs -
Position 133 : a vs -
Position 134 : c vs -
Position 135 : a vs -
Position 136 : g vs -
Position 137 : a vs -
Position 138 : a vs -
Position 139 : a vs -
Position 140 : a vs -
Position 141 : a vs -
Position 142 : a vs -
Position 143 : a vs -
Position 144 : g vs -
Position 145 : g vs a
Position 147 : a vs g
Position 151 : c vs t
Position 153 : g vs a
Position 154 : a vs t
Position 155 : a vs c
Position 160 : g vs c
Position 161 : a vs t
Position 164 : c vs -
Position 165 : c vs -
Position 166 : c vs -
Position 167 : t vs -
Position 168 : g vs -
Position 169 : t vs -
Position 170 : g vs -
Position 171 : t vs -
Position 172 : g vs -
```

```
Position 173 : a vs -
Position 174 : g vs -
Position 175 : a vs -
Position 176 : g vs -
Position 177 : a vs -
Position 178 : a vs -
Position 179 : a vs -
Position 180 : a vs -
Position 182 : a vs c
Position 183 : a vs g
Position 185 : g vs t
Position 189 : t vs g
Position 193 : c vs t
Position 195 : g vs c
Position 199 : c vs a
Position 202 : c vs t
Position 205 : - vs t
Position 206 : - vs t
Position 207 : - vs a
Position 208 : - vs a
Position 209 : - vs t
Position 210 : - vs g
Position 211 : - vs c
Position 213 : g vs a
Position 214 : c vs t
Position 215 : t vs g
Position 220 : g vs a
Position 225 : c vs t
Position 231 : a vs t
Position 232 : t vs g
Position 233 : a vs t
Position 235 : - vs c
Position 236 : - vs c
Position 237 : - vs a
Position 238 : - vs t
Position 239 : - vs c
Position 240 : - vs t
Position 241 : - vs c

Comparing: lcl|MF945602.1_cds_AXG51124.1_1 vs lcl|MF945601.1_cds_AXG51123.1_1
Position 1 : c vs -
Position 2 : t vs -
Position 3 : c vs -
Position 4 : a vs -
Position 5 : c vs -
Position 6 : t vs -
Position 7 : a vs -
Position 8 : a vs -
Position 9 : a vs -
Position 10 : g vs -
Position 11 : a vs -
Position 12 : c vs -
Position 13 : a vs -
Position 14 : g vs -
Position 15 : a vs -
Position 16 : a vs -
Position 17 : t vs -
Position 18 : g vs -
Position 19 : a vs -
Position 20 : a vs -
Position 21 : t vs -
Position 22 : g vs -
Position 23 : t vs -
Position 24 : a vs -
Position 25 : g vs -
Position 26 : a vs -
Position 27 : a vs -
```

```
Position 28 : a vs -
Position 29 : a vs -
Position 30 : g vs -
Position 31 : g vs -
Position 32 : c vs -
Position 33 : t vs -
Position 34 : g vs -
Position 35 : a vs -
Position 36 : a vs -
Position 37 : t vs -
Position 38 : t vs -
Position 39 : c vs -
Position 40 : t vs -
Position 41 : g vs -
Position 42 : t vs -
Position 43 : a vs -
Position 44 : a vs -
Position 45 : t vs -
Position 46 : a vs -
Position 47 : a vs -
Position 48 : a vs -

Comparing: lcl|MF945602.1_cds_AXG51124.1_1 vs lcl|KT120061.1_cds_ALF35866.1_1
Position 1 : c vs -
Position 2 : t vs -
Position 3 : c vs -
Position 4 : a vs -
Position 5 : c vs -
Position 6 : t vs -
Position 7 : a vs -
Position 8 : a vs -
Position 9 : a vs -
Position 10 : g vs -
Position 11 : a vs -
Position 12 : c vs -
Position 13 : a vs -
Position 14 : g vs -
Position 15 : a vs -
Position 16 : a vs -
Position 17 : t vs -
Position 18 : g vs -
Position 19 : a vs -
Position 20 : a vs -
Position 21 : t vs -
Position 22 : g vs -
Position 23 : t vs -
Position 24 : a vs -
Position 25 : g vs -
Position 26 : a vs -
Position 27 : a vs -
Position 28 : a vs -
Position 29 : a vs -
Position 30 : g vs -
Position 31 : g vs -
Position 32 : c vs -
Position 33 : t vs -
Position 34 : g vs -
Position 35 : a vs -
Position 36 : a vs -
Position 37 : t vs -
Position 38 : t vs -
Position 39 : c vs -
Position 40 : t vs -
Position 41 : g vs -
Position 42 : t vs -
Position 43 : a vs -
Position 44 : a vs -
```

Position 45 : t vs -
Position 46 : a vs -
Position 47 : a vs -
Position 48 : a vs -
Position 49 : a vs -
Position 50 : g vs -
Position 51 : c vs -
Position 52 : a vs -
Position 53 : a vs -
Position 54 : a vs -
Position 55 : c vs -
Position 56 : a vs -
Position 57 : g vs -
Position 58 : c vs -
Position 59 : c vs -
Position 60 : t vs -
Position 61 : g vs -
Position 62 : g vs -
Position 63 : c vs -
Position 64 : t vs -
Position 65 : t vs -
Position 66 : a vs -
Position 67 : g vs -
Position 68 : c vs -
Position 69 : a vs -
Position 70 : a vs -
Position 71 : g vs -
Position 72 : g vs -
Position 73 : a vs -
Position 74 : g vs -
Position 75 : c vs -
Position 76 : c vs -
Position 77 : a vs -
Position 78 : a vs -
Position 79 : c vs -
Position 80 : a vs -
Position 81 : t vs -
Position 82 : a vs -
Position 83 : a vs -
Position 84 : c vs -
Position 85 : a vs -
Position 86 : g vs -
Position 87 : a vs -
Position 88 : t vs -
Position 89 : g vs -
Position 90 : g vs -
Position 91 : g vs -
Position 92 : c vs -
Position 93 : t vs -
Position 94 : g vs -
Position 95 : g vs -
Position 96 : a vs -
Position 97 : a vs -
Position 98 : g vs -
Position 99 : t vs -
Position 100 : a vs -
Position 101 : a vs -
Position 102 : g vs -
Position 103 : g vs -
Position 104 : a vs -
Position 105 : a vs -
Position 106 : a vs -
Position 107 : c vs -
Position 108 : a vs -
Position 109 : t vs -
Position 110 : g vs -
Position 111 : t vs -

Position 112 : a vs -
Position 113 : a vs -
Position 114 : t vs -
Position 115 : g vs -
Position 116 : a vs -
Position 117 : t vs -
Position 118 : a vs -
Position 119 : g vs -
Position 120 : g vs -
Position 121 : c vs -
Position 122 : g vs -
Position 123 : g vs -
Position 124 : a vs -
Position 125 : c vs -
Position 126 : t vs -
Position 127 : c vs -
Position 128 : c vs -
Position 129 : c vs -
Position 130 : a vs -
Position 131 : g vs -
Position 132 : c vs -
Position 133 : a vs -
Position 134 : c vs -
Position 135 : a vs -
Position 136 : g vs -
Position 137 : a vs -
Position 138 : a vs -
Position 139 : a vs -
Position 140 : a vs -
Position 141 : a vs -
Position 142 : a vs -
Position 143 : a vs -
Position 144 : g vs -
Position 145 : g vs a
Position 147 : a vs g
Position 151 : c vs t
Position 153 : g vs a
Position 154 : a vs t
Position 155 : a vs c
Position 160 : g vs c
Position 161 : a vs t
Position 164 : c vs -
Position 165 : c vs -
Position 166 : c vs -
Position 167 : t vs -
Position 168 : g vs -
Position 169 : t vs -
Position 170 : g vs -
Position 171 : t vs -
Position 172 : g vs -
Position 173 : a vs -
Position 174 : g vs -
Position 175 : a vs -
Position 176 : g vs -
Position 177 : a vs -
Position 178 : a vs -
Position 179 : a vs -
Position 180 : a vs -
Position 182 : a vs c
Position 183 : a vs g
Position 185 : g vs t
Position 189 : t vs g
Position 193 : c vs t
Position 195 : g vs c
Position 199 : c vs a
Position 202 : c vs t
Position 205 : - vs t

```
Position 206 : - vs t
Position 207 : - vs a
Position 208 : - vs a
Position 209 : - vs t
Position 210 : - vs g
Position 211 : - vs c
Position 213 : g vs a
Position 214 : c vs t
Position 215 : t vs g
Position 220 : g vs a
Position 225 : c vs t
Position 231 : a vs t
Position 232 : t vs g
Position 233 : a vs t
Position 235 : t vs c
Position 236 : g vs c
Position 238 : a vs t
Position 239 : g vs c
Position 240 : a vs t
Position 241 : t vs c
Position 242 : g vs -
Position 243 : t vs -
Position 244 : t vs -
Position 245 : c vs -
Position 246 : c vs -
Position 247 : t vs -
Position 248 : t vs -
Position 249 : g vs -
Position 250 : g vs -
Position 251 : a vs -
Position 252 : t vs -
Position 253 : a vs -
Position 254 : a vs -
Position 255 : c vs -
Position 256 : a vs -
Position 257 : c vs -
Position 258 : t vs -
Position 259 : a vs -
Position 260 : a vs -
Position 261 : a vs -
Position 262 : t vs -
Position 263 : a vs -
Position 264 : g vs -
Position 265 : c vs -
Position 266 : a vs -
Position 267 : g vs -
Position 268 : c vs -
Position 269 : a vs -
Position 270 : t vs -
Position 271 : t vs -
Position 272 : c vs -
Position 273 : a vs -
Position 274 : g vs -
Position 275 : a vs -
Position 276 : a vs -
Position 277 : a vs -
Position 278 : g vs -

Comparing: lcl|MF945601.1_cds_AXG51123.1_1 vs lcl|KT120061.1_cds_ALF35866.1_1
Position 49 : a vs -
Position 50 : g vs -
Position 51 : c vs -
Position 52 : a vs -
Position 53 : a vs -
Position 54 : a vs -
Position 55 : c vs -
Position 56 : a vs -
```

Position 57 : g vs -
Position 58 : c vs -
Position 59 : c vs -
Position 60 : t vs -
Position 61 : g vs -
Position 62 : g vs -
Position 63 : c vs -
Position 64 : t vs -
Position 65 : t vs -
Position 66 : a vs -
Position 67 : g vs -
Position 68 : c vs -
Position 69 : a vs -
Position 70 : a vs -
Position 71 : g vs -
Position 72 : g vs -
Position 73 : a vs -
Position 74 : g vs -
Position 75 : c vs -
Position 76 : c vs -
Position 77 : a vs -
Position 78 : a vs -
Position 79 : c vs -
Position 80 : a vs -
Position 81 : t vs -
Position 82 : a vs -
Position 83 : a vs -
Position 84 : c vs -
Position 85 : a vs -
Position 86 : g vs -
Position 87 : a vs -
Position 88 : t vs -
Position 89 : g vs -
Position 90 : g vs -
Position 91 : g vs -
Position 92 : c vs -
Position 93 : t vs -
Position 94 : g vs -
Position 95 : g vs -
Position 96 : a vs -
Position 97 : a vs -
Position 98 : g vs -
Position 99 : t vs -
Position 100 : a vs -
Position 101 : a vs -
Position 102 : g vs -
Position 103 : g vs -
Position 104 : a vs -
Position 105 : a vs -
Position 106 : a vs -
Position 107 : c vs -
Position 108 : a vs -
Position 109 : t vs -
Position 110 : g vs -
Position 111 : t vs -
Position 112 : a vs -
Position 113 : a vs -
Position 114 : t vs -
Position 115 : g vs -
Position 116 : a vs -
Position 117 : t vs -
Position 118 : a vs -
Position 119 : g vs -
Position 120 : g vs -
Position 121 : c vs -
Position 122 : g vs -
Position 123 : g vs -

Position 124 : a vs -
Position 125 : c vs -
Position 126 : t vs -
Position 127 : c vs -
Position 128 : c vs -
Position 129 : c vs -
Position 130 : a vs -
Position 131 : g vs -
Position 132 : c vs -
Position 133 : a vs -
Position 134 : c vs -
Position 135 : a vs -
Position 136 : g vs -
Position 137 : a vs -
Position 138 : a vs -
Position 139 : a vs -
Position 140 : a vs -
Position 141 : a vs -
Position 142 : a vs -
Position 143 : a vs -
Position 144 : g vs -
Position 145 : g vs a
Position 147 : a vs g
Position 151 : c vs t
Position 153 : g vs a
Position 154 : a vs t
Position 155 : a vs c
Position 160 : g vs c
Position 161 : a vs t
Position 164 : c vs -
Position 165 : c vs -
Position 166 : c vs -
Position 167 : t vs -
Position 168 : g vs -
Position 169 : t vs -
Position 170 : g vs -
Position 171 : t vs -
Position 172 : g vs -
Position 173 : a vs -
Position 174 : g vs -
Position 175 : a vs -
Position 176 : g vs -
Position 177 : a vs -
Position 178 : a vs -
Position 179 : a vs -
Position 180 : a vs -
Position 182 : a vs c
Position 183 : a vs g
Position 185 : g vs t
Position 189 : t vs g
Position 193 : c vs t
Position 195 : g vs c
Position 199 : c vs a
Position 202 : c vs t
Position 205 : - vs t
Position 206 : - vs t
Position 207 : - vs a
Position 208 : - vs a
Position 209 : - vs t
Position 210 : - vs g
Position 211 : - vs c
Position 213 : g vs a
Position 214 : c vs t
Position 215 : t vs g
Position 220 : g vs a
Position 225 : c vs t
Position 231 : a vs t

```
Position 232 : t vs g
Position 233 : a vs t
Position 235 : t vs c
Position 236 : g vs c
Position 238 : a vs t
Position 239 : g vs c
Position 240 : a vs t
Position 241 : t vs c
Position 242 : g vs -
Position 243 : t vs -
Position 244 : t vs -
Position 245 : c vs -
Position 246 : c vs -
Position 247 : t vs -
Position 248 : t vs -
Position 249 : g vs -
Position 250 : g vs -
Position 251 : a vs -
Position 252 : t vs -
Position 253 : a vs -
Position 254 : a vs -
Position 255 : c vs -
Position 256 : a vs -
Position 257 : c vs -
Position 258 : t vs -
Position 259 : a vs -
Position 260 : a vs -
Position 261 : a vs -
Position 262 : t vs -
Position 263 : a vs -
Position 264 : g vs -
Position 265 : c vs -
Position 266 : a vs -
Position 267 : g vs -
Position 268 : c vs -
Position 269 : a vs -
Position 270 : t vs -
Position 271 : t vs -
Position 272 : c vs -
Position 273 : a vs -
Position 274 : g vs -
Position 275 : a vs -
Position 276 : a vs -
Position 277 : a vs -
Position 278 : g vs -
```

3. Identifying SNPs

Using MF945608.1 as the reference (wild type), almost all of the sequences showed no SNPs. Only one sequence, MF945606.1, had a single SNP at position 194, where the base changes from **a** to **g**. All the other sequences in the MF94560x group match the reference at every position where a real base is present. The only sequence with many SNPs is KT120061.1, which shows a list of 30 base substitutions across the gene. These include changes such as **g** to **a** at position 145, **a** to **g** at 147, c to t at 151, and many others, all the way through positions 235 to 241.

In [38]:
```
snps_per_sequence(records2)
```

```
SNP counts relative to reference: lcl|MF945608.1_cds_AXG51130.1_1

lcl|MF945608.1_cds_AXG51130.1_1 → 0 SNPs
lcl|MF945607.1_cds_AXG51129.1_1 → 0 SNPs
lcl|MF945606.1_cds_AXG51128.1_1 → 1 SNPs
lcl|MF945605.1_cds_AXG51127.1_1 → 0 SNPs
lcl|MF945604.1_cds_AXG51126.1_1 → 0 SNPs
lcl|MF945603.1_cds_AXG51125.1_1 → 0 SNPs
lcl|MF945602.1_cds_AXG51124.1_1 → 0 SNPs
lcl|MF945601.1_cds_AXG51123.1_1 → 0 SNPs
lcl|KT120061.1_cds_ALF35866.1_1 → 30 SNPs
```

In [39]:
```
identify_snps(records2)
```

```
Reference sequence: lcl|MF945608.1_cds_AXG51130.1_1

lcl|MF945608.1_cds_AXG51130.1_1

lcl|MF945607.1_cds_AXG51129.1_1

lcl|MF945606.1_cds_AXG51128.1_1
  Position 194 : a → g

lcl|MF945605.1_cds_AXG51127.1_1

lcl|MF945604.1_cds_AXG51126.1_1

lcl|MF945603.1_cds_AXG51125.1_1

lcl|MF945602.1_cds_AXG51124.1_1

lcl|MF945601.1_cds_AXG51123.1_1

lcl|KT120061.1_cds_ALF35866.1_1
  Position 145 : g → a
  Position 147 : a → g
  Position 151 : c → t
  Position 153 : g → a
  Position 154 : a → t
  Position 155 : a → c
  Position 160 : g → c
  Position 161 : a → t
  Position 182 : a → c
  Position 183 : a → g
  Position 185 : g → t
  Position 189 : t → g
  Position 193 : c → t
  Position 195 : g → c
  Position 199 : c → a
  Position 202 : c → t
  Position 213 : g → a
  Position 214 : c → t
  Position 215 : t → g
  Position 220 : g → a
  Position 225 : c → t
  Position 231 : a → t
  Position 232 : t → g
  Position 233 : a → t
  Position 235 : t → c
  Position 236 : g → c
  Position 238 : a → t
  Position 239 : g → c
  Position 240 : a → t
  Position 241 : t → c
```

4. Identifying INDELs

Using MF945608.1 as the reference, most sequences show no structural changes. MF945607.1, MF945606.1, and MF945602.1 presented 0 INDELs, meaning they have no deletions or insertions relative to the reference. Two sequences, MF945605.1 and MF945604.1, both show the same pattern of 12 deletions at positions 1–12, where the bases **ctcactaaagac** are missing from the start of the gene. The sequence MF945603.1 contains 44 deletions, all grouped together at the end of the gene (positions 235–278), showing that the entire tail of the sequence has been deleted. MF945601.1 has 48 deletions, starting from position 1 and continuing through position 48, meaning almost the whole beginning of the gene is deleted. The most extreme case is KT120061.1, which contains 205 INDELs, including long stretches of deletions across the entire gene and a small cluster of insertions at positions 205–211. These insertions add the bases ttaatgc, which do not appear in the reference.

In [40]:
```
identify_indels(records2)
```

INDELs relative to reference: lcl|MF945608.1_cds_AXG51130.1_1

lcl|MF945608.1_cds_AXG51130.1_1 (0 INDELs)

lcl|MF945607.1_cds_AXG51129.1_1 (0 INDELs)

lcl|MF945606.1_cds_AXG51128.1_1 (0 INDELs)

lcl|MF945605.1_cds_AXG51127.1_1 (12 INDELs)
  Position 1: deletion (c)
  Position 2: deletion (t)
  Position 3: deletion (c)
  Position 4: deletion (a)
  Position 5: deletion (c)
  Position 6: deletion (t)
  Position 7: deletion (a)
  Position 8: deletion (a)
  Position 9: deletion (a)
  Position 10: deletion (g)
  Position 11: deletion (a)
  Position 12: deletion (c)

lcl|MF945604.1_cds_AXG51126.1_1 (12 INDELs)
  Position 1: deletion (c)
  Position 2: deletion (t)
  Position 3: deletion (c)
  Position 4: deletion (a)
  Position 5: deletion (c)
  Position 6: deletion (t)
  Position 7: deletion (a)
  Position 8: deletion (a)
  Position 9: deletion (a)
  Position 10: deletion (g)
  Position 11: deletion (a)
  Position 12: deletion (c)

lcl|MF945603.1_cds_AXG51125.1_1 (44 INDELs)
  Position 235: deletion (t)
  Position 236: deletion (g)
  Position 237: deletion (a)
  Position 238: deletion (a)
  Position 239: deletion (g)
  Position 240: deletion (a)
  Position 241: deletion (t)
  Position 242: deletion (g)
  Position 243: deletion (t)
  Position 244: deletion (t)
  Position 245: deletion (c)
  Position 246: deletion (c)
  Position 247: deletion (t)
  Position 248: deletion (t)
  Position 249: deletion (g)
  Position 250: deletion (g)
  Position 251: deletion (a)
  Position 252: deletion (t)
  Position 253: deletion (a)
  Position 254: deletion (a)
  Position 255: deletion (c)
  Position 256: deletion (a)
  Position 257: deletion (c)
  Position 258: deletion (t)
  Position 259: deletion (a)
  Position 260: deletion (a)
  Position 261: deletion (a)
  Position 262: deletion (t)
  Position 263: deletion (a)
  Position 264: deletion (g)

```
  Position 265: deletion (c)
  Position 266: deletion (a)
  Position 267: deletion (g)
  Position 268: deletion (c)
  Position 269: deletion (a)
  Position 270: deletion (t)
  Position 271: deletion (t)
  Position 272: deletion (c)
  Position 273: deletion (a)
  Position 274: deletion (g)
  Position 275: deletion (a)
  Position 276: deletion (a)
  Position 277: deletion (a)
  Position 278: deletion (g)

lcl|MF945602.1_cds_AXG51124.1_1 (0 INDELs)

lcl|MF945601.1_cds_AXG51123.1_1 (48 INDELs)
  Position 1: deletion (c)
  Position 2: deletion (t)
  Position 3: deletion (c)
  Position 4: deletion (a)
  Position 5: deletion (c)
  Position 6: deletion (t)
  Position 7: deletion (a)
  Position 8: deletion (a)
  Position 9: deletion (a)
  Position 10: deletion (g)
  Position 11: deletion (a)
  Position 12: deletion (c)
  Position 13: deletion (a)
  Position 14: deletion (g)
  Position 15: deletion (a)
  Position 16: deletion (a)
  Position 17: deletion (t)
  Position 18: deletion (g)
  Position 19: deletion (a)
  Position 20: deletion (a)
  Position 21: deletion (t)
  Position 22: deletion (g)
  Position 23: deletion (t)
  Position 24: deletion (a)
  Position 25: deletion (g)
  Position 26: deletion (a)
  Position 27: deletion (a)
  Position 28: deletion (a)
  Position 29: deletion (a)
  Position 30: deletion (g)
  Position 31: deletion (g)
  Position 32: deletion (c)
  Position 33: deletion (t)
  Position 34: deletion (g)
  Position 35: deletion (a)
  Position 36: deletion (a)
  Position 37: deletion (t)
  Position 38: deletion (t)
  Position 39: deletion (c)
  Position 40: deletion (t)
  Position 41: deletion (g)
  Position 42: deletion (t)
  Position 43: deletion (a)
  Position 44: deletion (a)
  Position 45: deletion (t)
  Position 46: deletion (a)
  Position 47: deletion (a)
  Position 48: deletion (a)
```

lcl|KT120061.1_cds_ALF35866.1_1 (205 INDELs)
  Position 1: deletion (c)
  Position 2: deletion (t)
  Position 3: deletion (c)
  Position 4: deletion (a)
  Position 5: deletion (c)
  Position 6: deletion (t)
  Position 7: deletion (a)
  Position 8: deletion (a)
  Position 9: deletion (a)
  Position 10: deletion (g)
  Position 11: deletion (a)
  Position 12: deletion (c)
  Position 13: deletion (a)
  Position 14: deletion (g)
  Position 15: deletion (a)
  Position 16: deletion (a)
  Position 17: deletion (t)
  Position 18: deletion (g)
  Position 19: deletion (a)
  Position 20: deletion (a)
  Position 21: deletion (t)
  Position 22: deletion (g)
  Position 23: deletion (t)
  Position 24: deletion (a)
  Position 25: deletion (g)
  Position 26: deletion (a)
  Position 27: deletion (a)
  Position 28: deletion (a)
  Position 29: deletion (a)
  Position 30: deletion (g)
  Position 31: deletion (g)
  Position 32: deletion (c)
  Position 33: deletion (t)
  Position 34: deletion (g)
  Position 35: deletion (a)
  Position 36: deletion (a)
  Position 37: deletion (t)
  Position 38: deletion (t)
  Position 39: deletion (c)
  Position 40: deletion (t)
  Position 41: deletion (g)
  Position 42: deletion (t)
  Position 43: deletion (a)
  Position 44: deletion (a)
  Position 45: deletion (t)
  Position 46: deletion (a)
  Position 47: deletion (a)
  Position 48: deletion (a)
  Position 49: deletion (a)
  Position 50: deletion (g)
  Position 51: deletion (c)
  Position 52: deletion (a)
  Position 53: deletion (a)
  Position 54: deletion (a)
  Position 55: deletion (c)
  Position 56: deletion (a)
  Position 57: deletion (g)
  Position 58: deletion (c)
  Position 59: deletion (c)
  Position 60: deletion (t)
  Position 61: deletion (g)
  Position 62: deletion (g)
  Position 63: deletion (c)
  Position 64: deletion (t)
  Position 65: deletion (t)
  Position 66: deletion (a)

Position 67: deletion (g)
Position 68: deletion (c)
Position 69: deletion (a)
Position 70: deletion (a)
Position 71: deletion (g)
Position 72: deletion (g)
Position 73: deletion (a)
Position 74: deletion (g)
Position 75: deletion (c)
Position 76: deletion (c)
Position 77: deletion (a)
Position 78: deletion (a)
Position 79: deletion (c)
Position 80: deletion (a)
Position 81: deletion (t)
Position 82: deletion (a)
Position 83: deletion (a)
Position 84: deletion (c)
Position 85: deletion (a)
Position 86: deletion (g)
Position 87: deletion (a)
Position 88: deletion (t)
Position 89: deletion (g)
Position 90: deletion (g)
Position 91: deletion (g)
Position 92: deletion (c)
Position 93: deletion (t)
Position 94: deletion (g)
Position 95: deletion (g)
Position 96: deletion (a)
Position 97: deletion (a)
Position 98: deletion (g)
Position 99: deletion (t)
Position 100: deletion (a)
Position 101: deletion (a)
Position 102: deletion (g)
Position 103: deletion (g)
Position 104: deletion (a)
Position 105: deletion (a)
Position 106: deletion (a)
Position 107: deletion (c)
Position 108: deletion (a)
Position 109: deletion (t)
Position 110: deletion (g)
Position 111: deletion (t)
Position 112: deletion (a)
Position 113: deletion (a)
Position 114: deletion (t)
Position 115: deletion (g)
Position 116: deletion (a)
Position 117: deletion (t)
Position 118: deletion (a)
Position 119: deletion (g)
Position 120: deletion (g)
Position 121: deletion (c)
Position 122: deletion (g)
Position 123: deletion (g)
Position 124: deletion (a)
Position 125: deletion (c)
Position 126: deletion (t)
Position 127: deletion (c)
Position 128: deletion (c)
Position 129: deletion (c)
Position 130: deletion (a)
Position 131: deletion (g)
Position 132: deletion (c)
Position 133: deletion (a)

Position 134: deletion (c)
Position 135: deletion (a)
Position 136: deletion (g)
Position 137: deletion (a)
Position 138: deletion (a)
Position 139: deletion (a)
Position 140: deletion (a)
Position 141: deletion (a)
Position 142: deletion (a)
Position 143: deletion (a)
Position 144: deletion (g)
Position 164: deletion (c)
Position 165: deletion (c)
Position 166: deletion (c)
Position 167: deletion (t)
Position 168: deletion (g)
Position 169: deletion (t)
Position 170: deletion (g)
Position 171: deletion (t)
Position 172: deletion (g)
Position 173: deletion (a)
Position 174: deletion (g)
Position 175: deletion (a)
Position 176: deletion (g)
Position 177: deletion (a)
Position 178: deletion (a)
Position 179: deletion (a)
Position 180: deletion (a)
Position 205: insertion (t)
Position 206: insertion (t)
Position 207: insertion (a)
Position 208: insertion (a)
Position 209: insertion (t)
Position 210: insertion (g)
Position 211: insertion (c)
Position 242: deletion (g)
Position 243: deletion (t)
Position 244: deletion (t)
Position 245: deletion (c)
Position 246: deletion (c)
Position 247: deletion (t)
Position 248: deletion (t)
Position 249: deletion (g)
Position 250: deletion (g)
Position 251: deletion (a)
Position 252: deletion (t)
Position 253: deletion (a)
Position 254: deletion (a)
Position 255: deletion (c)
Position 256: deletion (a)
Position 257: deletion (c)
Position 258: deletion (t)
Position 259: deletion (a)
Position 260: deletion (a)
Position 261: deletion (a)
Position 262: deletion (t)
Position 263: deletion (a)
Position 264: deletion (g)
Position 265: deletion (c)
Position 266: deletion (a)
Position 267: deletion (g)
Position 268: deletion (c)
Position 269: deletion (a)
Position 270: deletion (t)
Position 271: deletion (t)
Position 272: deletion (c)
Position 273: deletion (a)

```
Position 274: deletion (g)
Position 275: deletion (a)
Position 276: deletion (a)
Position 277: deletion (a)
Position 278: deletion (g)
```

DISCUSSION

The BRCA1 mRNA isolates (DATA SET 1) form a highly conserved group, with almost no substantial structural variation. Across all sequences in this dataset, only 1 SNP appears, and a small number of deletions or insertions. Some sequences presented short insertions at the 5′ end (positions 1–3), while others contain small deletions near the 3′ end (positions 238–241). These INDELs are limited in size and occur in restricted regions rather than being distributed across the gene, but some of these INDELs involve 1–3 nucleotides; they could give them the potential to alter the reading frame (Quann et al. 2015). However, the overall pattern suggests that the BRCA1 mRNA isolates remain intact, with only minor disruptions, suggesting that the coding region may be intact in every sequence, and the reading frame is preserved. Probably the protein structure that could arise would not be disrupted (futher analisys are required to test this hypothesis) (Quann et al. 2015). The small number of SNPs observed suggests minor substitutions that do not affect the overall gene structure, suggesting that this indicates that the mRNA isolates represent a stable, full-length BRCA1 coding region with very limited genetic variation. This strong conservation could be justified by the fact that BRCA1 has an unusually long coding region of more than 5.5 kb (78% of the mRNA length), which is highly conserved and also because BRCA1 is a critical regulator of genomic integrity undergoing strong selective pressure to remain intact (Quann et al. 2015). Also, Nepomuceno et al. (2022) demonstrated in a functional assay that “amino acid substitutions, deletions or insertions in the disordered region do not significantly impact activity and are not likely to constitute pathogenic alleles. For this specific data set, we could suggest that DATA set 1 represents an almost normal structure of the BRCA1 gene sequences.

On the other hand, the partial BRCA1 isolates in DATA SET 2 showed a pattern that is different from the BRCA1 mRNA isolates. Instead of forming a uniform group, they present a stratification into a small conserved subgroup. The higher number of indels suggests that these genes represented by those sequences would possibly be inactive and possibly not able to produce a functional protein. This pattern is expected when analysing incomplete gene fragments, because once a sequence loses its ability to encode a functional protein, it is no longer maintained by purifying selection and therefore accumulates mutations, deletions, and insertions at a much higher rate, which indicates that the dataset contains sequences at different stages of degradation(Zhang 2003). Studies show that in breast cancer, BRCA1-deficient cells accumulate a lot of indels that include long deletion specially when BRCA1-dependent repair pathways fail. For example, Lal et al. 2019 performed a comprehensive genomic analysis of breast tumours carrying BRCA1 mutations, integrating whole-genome sequencing and structural variant calling, where they found that BRCA1-mutated tumours show an elevated burden of structural variation, especially deletions, duplications, and chromosomal translocations. Additionally, Yu et al. 2025) suggest that Alu elements misalign during repair and generate large exon-spanning deletions that inactivate BRCA1. Taking into consideration these statements, the high variability in DATA SET two suggests that those files represent different parts or BRCA1 or different stages of degradation of this gene in a tumorigenesis scenario, but further analysis, such as functional analysis or mutational signatures, would be necessary.

key limitation of this analysis is the absence of species information for the BRCA1-related sequences. BRCA1 is highly conserved across mammals, but cross-species comparisons can produce lower similarity scores and longer phylogenetic branch lengths (Quann et al. 2015). For example, the extreme outlier KT120061 may reflect species-level differences rather than genuine structural

instability within the gene.

limitations

1. No functional or clinical correlation
2. low number of studies using similar tools for results comparison
3. Small dataset sizes, which could compromise the statistical power
4. No experimental validation
5. Unknown Species Origin of the Sequences

conclusions

Two analysed data sets of the BRCA1 (mRNA isolates and partial gene isolates) revealed clear differences in structural integrity and mutational patterns between the sequences. The analysis showed that most isolates in the BRCA1 mRNA isolates showed no structural changes relative to the reference, while the partial isolates presented extensive deletions, terminal losses and some substitutions that could consequently inactivate the gene and affect the protein translation, suggesting that these variants could be biologically relevant for future analyses to understand the structural mechanisms underlying BRCA1 disruption, including Alu-mediated recombination and replication-stress-induced.

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BRCA1 structural variation and breast cancer susceptibility. Genes & Genomics 48 (1), 1–13. Zhang, J. (2003) Evolution by gene duplication: an update. Trends in Ecology & Evolution 18 (6), 292–298.